

Lecture 5: Hypothesis Testing

L2 - Statistics

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Today's lecture

The big question: How do we test whether our data provides evidence for or against a claim about the population?

1. Introduction to hypothesis testing
2. The hypothesis testing framework
3. One-sample tests
4. Two-sample tests
5. Paired tests
6. Chi-square test of independence
7. Randomization methods
8. Statistical power and sample size
9. Practical guidelines

Introduction to hypothesis testing

From confidence intervals to hypothesis tests

Last lecture: We learned how to construct confidence intervals to quantify uncertainty about population parameters.

Example: Life satisfaction in France

We estimated that the average life satisfaction in France is **7.0 (95% CI: 6.9 to 7.1)** based on European Social Survey data.

This lecture: Instead of estimating a range of plausible values, we often want to test specific claims about the population.

Research questions

- Is the average life satisfaction in France different from 7?
- Do men and women have different levels of life satisfaction?
- Is trust in parliament higher than trust in politicians?
- Does interpersonal trust really vary by education level?

What is a hypothesis test?

A **hypothesis test** is a statistical method for making decisions about population parameters based on sample data.

The logic:

1. Start with a claim (hypothesis) about the population
2. Collect sample data
3. Calculate how **compatible** the data is with the claim
4. Make a decision: *reject the claim* or *fail to reject it*

Important

We never “prove” a hypothesis is true. We only assess whether the data provides sufficient evidence against a claimed value.

A simple example

Suppose a politician claims that **60% of French adults trust parliament.**

You survey 100 randomly selected adults and find that **45% trust parliament.**

Questions:

- Is this evidence against the politician's claim?
- Could we observe 45% just by chance if the true proportion is 60%?
- How unlikely would our result need to be before we reject the claim?

Hypothesis testing provides a formal framework to answer these questions.

Connection to confidence intervals

Hypothesis testing and confidence intervals are **two sides of the same coin**.

Equivalence

- If a 95% confidence interval for μ is $(6.9, 7.1)$, then we would **reject** any hypothesis that μ equals a value outside this interval (e.g., $\mu = 6.5$) at the 5% significance level.
- If a hypothesized value μ_0 falls inside the 95% CI, we would **fail to reject** the hypothesis that $\mu = \mu_0$ at the 5% level.

Today: We'll learn the formal hypothesis testing framework, which gives us more flexibility:

- One-sided vs two-sided tests
- Comparing groups
- Testing relationships between variables

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The hypothesis testing framework

The six steps of hypothesis testing

Every hypothesis test follows the same structure:

1. **State the hypotheses:** Define null (H_0) and alternative (H_A) hypotheses
2. **Choose significance level:** Set significance level α (typically 0.05)
3. **Calculate test statistic:** Compute from sample data
4. **Find the p-value:** Probability of observing our result (or more extreme) if H_0 is true
5. **Make a decision:** Reject H_0 if p-value $< \alpha$
6. **Interpret in context:** State conclusion in plain language

Step 1: State the hypotheses

Every hypothesis test involves two competing hypotheses:

! Definitions

Null hypothesis (H_0): The claim we are testing, typically representing “no effect” or “no difference”

Alternative hypothesis (H_A or H_1): What we conclude if we reject H_0

Key principle: H_0 always contains an equality ($=$, $\leq$, or $\geq$)

i Example: Life satisfaction

- $H_0 : \mu = 7$ (average life satisfaction equals 7)
- $H_A : \mu \neq 7$ (average life satisfaction differs from 7)

One-sided vs two-sided tests

The alternative hypothesis can be:

Two-sided (non-directional):

$$H_0 : \mu = \mu_0 \quad \text{vs} \quad H_A : \mu \neq \mu_0$$

Used when we care about differences in *either* direction.

One-sided (directional):

$$H_0 : \mu \leq \mu_0 \quad \text{vs} \quad H_A : \mu > \mu_0$$

or

$$H_0 : \mu \geq \mu_0 \quad \text{vs} \quad H_A : \mu < \mu_0$$

Used when we only care about differences in *one* direction.

Step 2: Choose significance level (α)

The **significance level** (α) is the probability threshold for rejecting H_0 .

Common choices:

- $\alpha = 0.05$ (5% level) – most common
- $\alpha = 0.01$ (1% level) – more stringent
- $\alpha = 0.10$ (10% level) – more lenient

Interpretation:

$\alpha = 0.05$ means we're willing to reject H_0 when there's only a 5% chance of observing data this extreme if H_0 were true.

Note

The significance level α represents the **maximum probability of a Type I error** we're willing to tolerate (more on this soon!).

Step 3: Calculate the test statistic

A **test statistic** measures how far our sample estimate is from the hypothesized value, in units of standard errors.

For a population mean:

$$t = \frac{\bar{x} - \mu_0}{SE(\bar{x})} = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

where:

- $\bar{x}$ = sample mean
- μ_0 = hypothesized population mean
- s = sample standard deviation
- n = sample size

Interpretation: The test statistic tells us “how many standard errors away” our sample estimate is from the hypothesized value.

Step 4: Calculate the p-value

! Definition

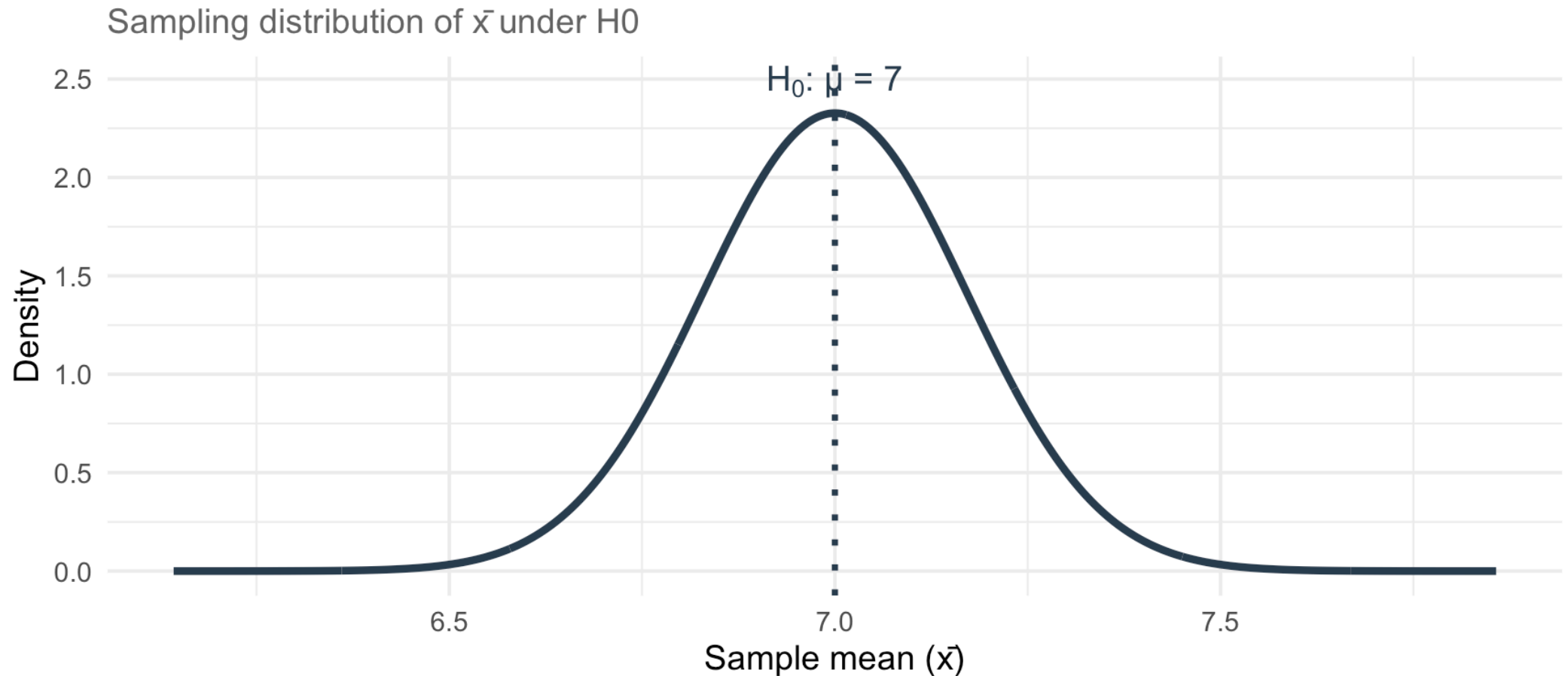
The **p-value** is the probability of observing a test statistic as extreme as (or more extreme than) what we actually observed, *assuming the null hypothesis H_0 is true*.

Example setup — testing whether mean life satisfaction in France equals 7:

- $H_0 : \mu = 7$ vs $H_A : \mu \neq 7$
- Significance level: 5%
- Observed sample mean: $\bar{x} = 7.4$, with $s = 2.1$, $n = 150$
- Test statistic: $t = \frac{7.4 - 7}{2.1/\sqrt{150}} \approx 2.33$

Step 4: The null distribution

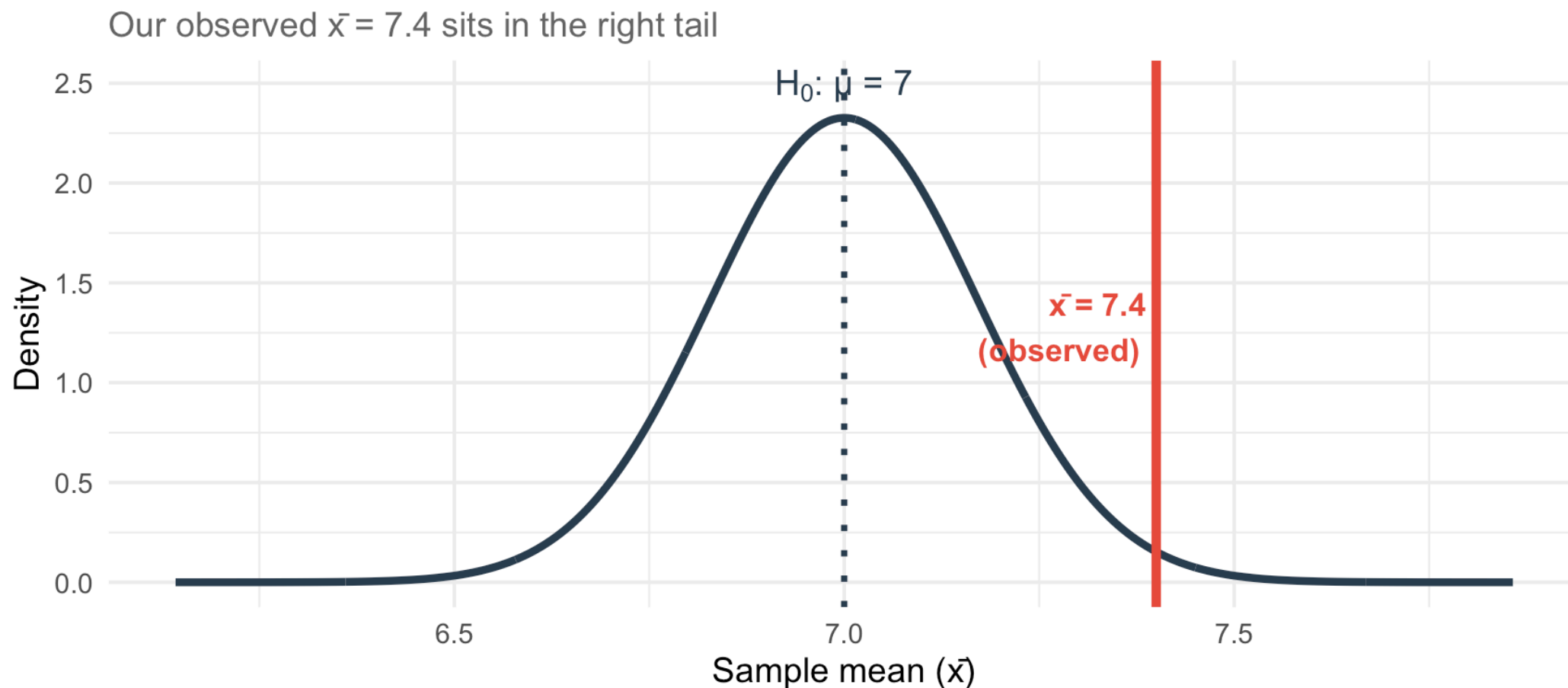
Under H_0 , if $\mu = 7$, repeated samples of size $n = 150$ would give sample means distributed as:



This is the distribution of $\bar{x}$ we'd expect to see **if the null were true**.

Step 4: Where did our sample fall?

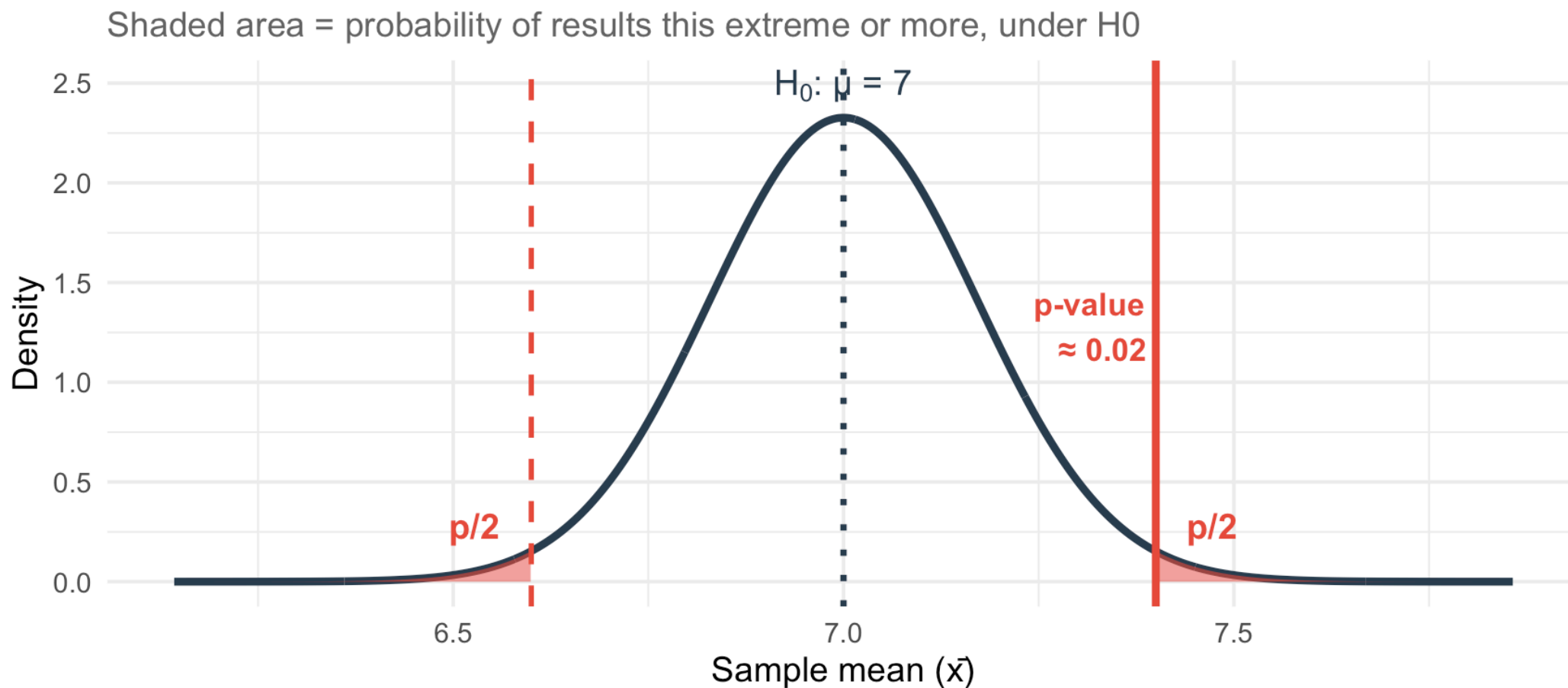
We observed $\bar{x} = 7.4$. How does it sit relative to the null distribution?



Our result looks unusual under H_0 — but how unusual exactly?

Step 4: The p-value area

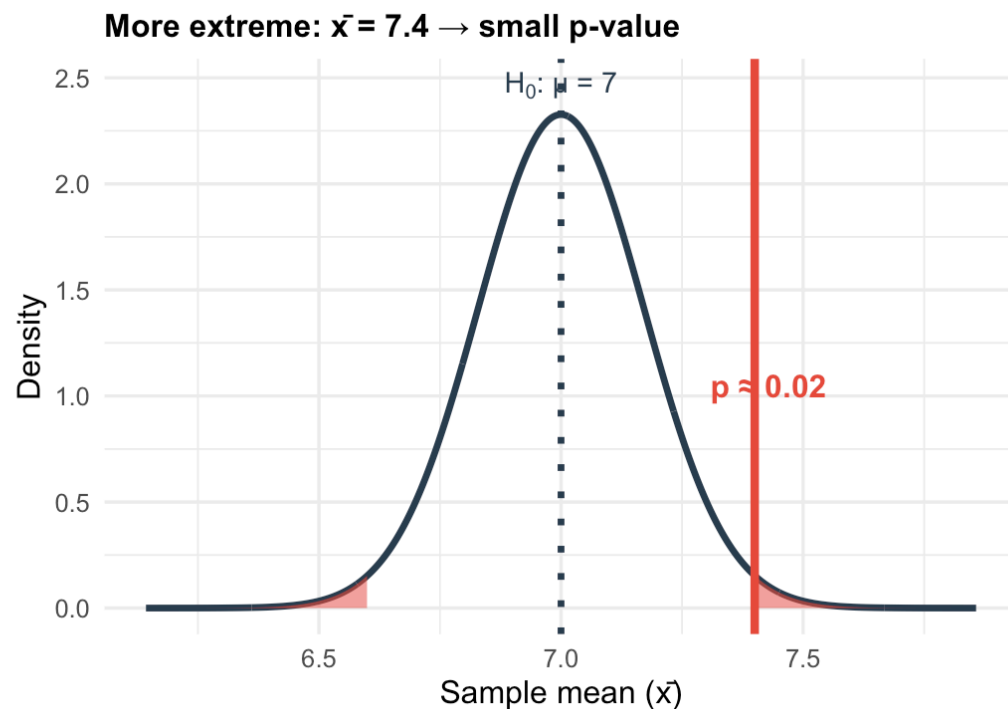
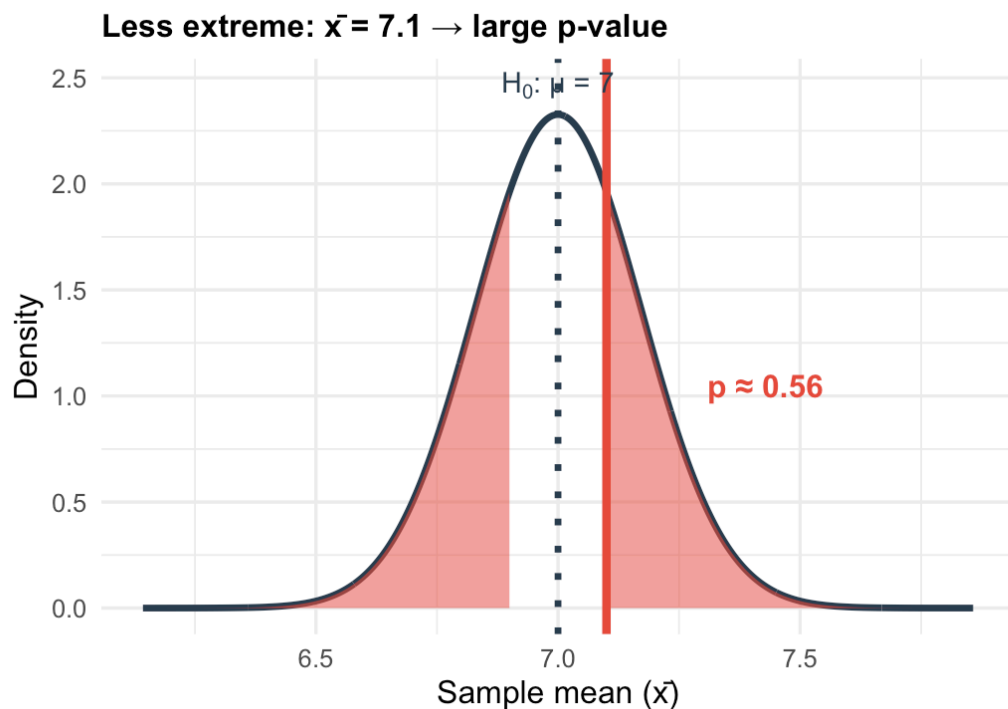
The p-value = $P(|\bar{x} - 7| \geq 0.4 \mid H_0)$ – the shaded area in **both tails**:



p-value ≈ 0.02 – the data are unlikely under H_0 . We reject H_0 at the 5% level. For a two-tailed test, the p-value is computed by: `2 * (1 - pnorm(abs(obs_value - mu0) / se))`

Step 4: How the p-value changes with the test statistic

The further the observed $\bar{x}$ from μ_0 , the smaller the p-value:



Interpreting p-values

! What the p-value is (and isn't)

The p-value IS:

- The probability of obtaining results as extreme as observed, *assuming H_0 is true*
- A measure of compatibility between data and H_0

The p-value is NOT:

- The probability that H_0 is true
- The probability that the results occurred by chance
- The probability that you've made a wrong decision

Common interpretations:

- **Small p-value** ($< \alpha$): Data is incompatible with $H_0 \rightarrow$ reject H_0
- **Large p-value** ($\geq \alpha$): Data is **compatible** with $H_0 \rightarrow$ fail to reject H_0

! Warning

"Fail to reject H_0 " $\neq$ "Accept H_0 " or "Prove H_0 is true"

We simply don't have sufficient evidence against H_0 .

Step 5: Make a decision

Compare the p-value to the significance level α :

Decision rule:

- If p-value $< \alpha$: **Reject H_0** (results are “statistically significant”)
- If **p-value $\geq \alpha$** : **Fail to reject H_0** (results are “not statistically significant”)

 The 0.05 threshold is arbitrary!

The conventional threshold of $\alpha = 0.05$ is **not a magical number**. It's a historical convention introduced by [Fisher \(1925\)](#) — chosen as a convenient round number, not for any deep statistical reason.

- There's nothing special about 5% vs 4.9% or 5.1%
- Results with $p = 0.049$ and $p = 0.051$ are not fundamentally different
- **Don't treat 0.05 as a bright line** — report actual p-values and effect sizes!

Step 6: Interpret in context

Always translate your statistical decision back into the real-world context.

Bad interpretations:

- ❌ “We reject the null hypothesis”
- ❌ “The p-value is 0.03”
- ❌ “The result is significant”

Good interpretations:

- ✅ “The data provide strong evidence that average life satisfaction in France differs from 7 ($p = 0.03$)”
- ✅ “We found a statistically significant difference in life satisfaction between men and women ($p = 0.02$)”
- ✅ “There is insufficient evidence to conclude that trust in parliament differs from trust in politicians ($p = 0.18$)”

Your turn! #1

12:00

A government report states that French adults have an average interest in politics of 5 on a 0 (*Not at all interested*) – 10 (*Very interested*) scale. You survey a random sample of 50 French university students and find: $\bar{x} = 5.8$, with standard deviation $s = 2.3$.

Work through all six steps to test whether students differ from the general population:

Type I and Type II errors

In hypothesis testing, we can make two types of errors:

	H_0 is true	H_0 is false (H_a is true)
Reject H_0	✗ Type I error False Positive Probability = α	✓ Correct decision Power Probability = $1 - \beta$
Fail to reject H_0	✓ Correct decision Probability = $1 - \alpha$	✗ Type II error False Negative Probability = β

! Definitions

- **Type I error (α)**: Rejecting H_0 when it's actually true (false positive)
- **Type II error (β)**: Failing to reject H_0 when it's actually false (false negative)
- **Power ($1 - \beta$)**: Probability of correctly rejecting a false H_0

Understanding Type I and Type II errors

Medical test analogy

Think of a medical test for a disease:

- **Type I error:** Test says you have the disease when you don't (false positive)
- **Type II error:** Test says you don't have the disease when you do (false negative)

In statistical testing:

- We control Type I error by choosing α (typically 0.05)
- We **cannot directly control Type II error** without knowing the true parameter value
- Type II error depends on: sample size, effect size, and α

The trade-off:

- Decreasing α (e.g., from 0.05 to 0.01) reduces Type I error
- But it **increases** Type II error (harder to detect real effects)

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One-sample tests

One-sample test for means

Goal: Test whether a population mean μ equals a specific hypothesized value μ_0 .

Hypotheses:

$$H_0 : \mu = \mu_0 \quad \text{vs} \quad H_A : \mu \neq \mu_0$$

Test statistic:

$$T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}$$

Under H_0 : $T \sim t_{n-1}$

i If σ is known

If we know the true population SD σ , we can use the standard normal distribution $N(0, 1)$. Since in practice we **estimate** σ with the sample SD s , we use the t-distribution.

P-value (two-sided): $P(|T| \geq |t_{\text{obs}}| = \left| \frac{\bar{x} - \mu_0}{s/\sqrt{n}} \right|)$ where $T \sim t_{n-1}$

Example: Trust in parliament

Research question: Is the average trust in parliament among French adults equal to 5 (the midpoint of the 0-10 scale)?

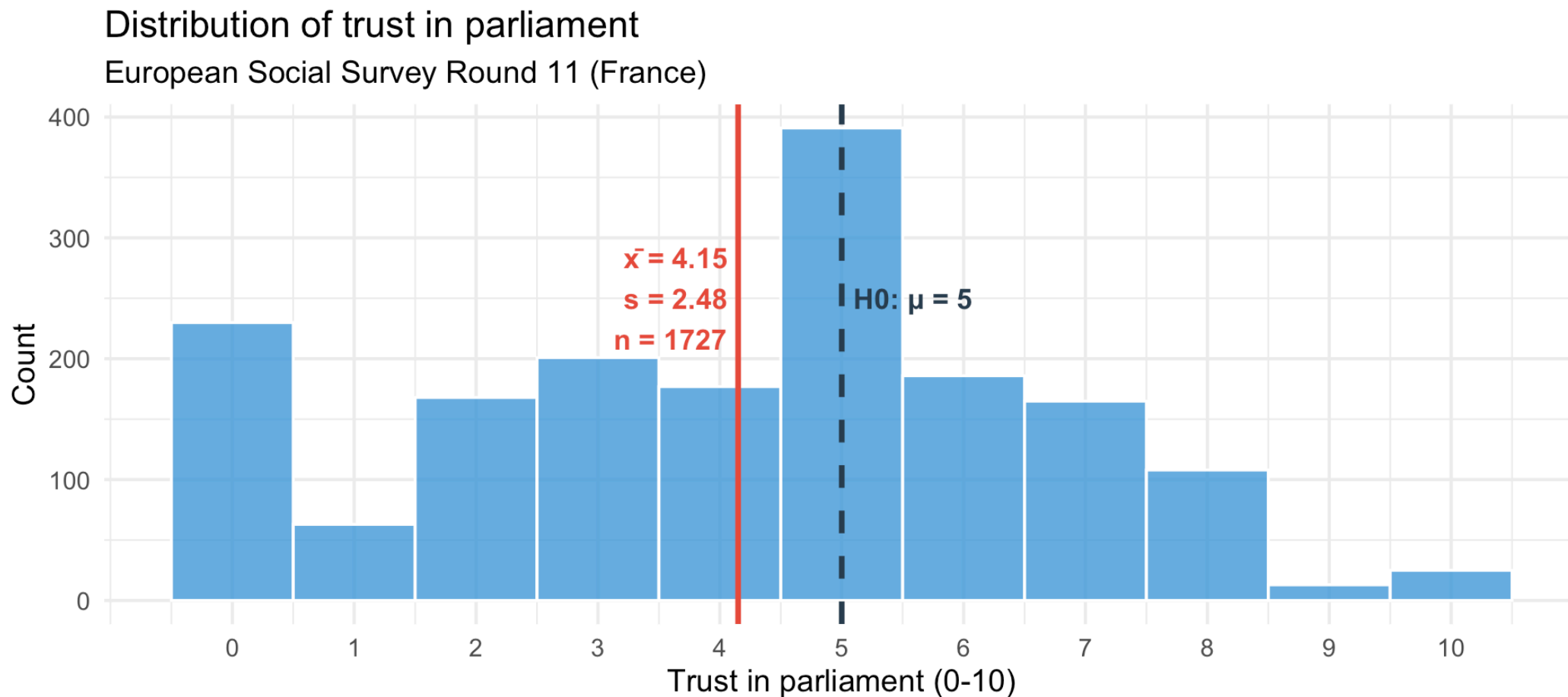
```
1 # Load ESS data
2 ess_france <- read.csv("https://www.dropbox.com/scl/fi/25h4lezq3zuzla94ejmqq/ess_france_11ed.csv?rlke
3
4 # Clean trust in parliament variable
5 ess_trust_parl <- ess_france |>
6   filter(trstprl %in% 0:10) |>
7   select(trstprl)
8
9 # Summary statistics
10 n <- nrow(ess_trust_parl)
11 xbar <- mean(ess_trust_parl$trstprl)
12 s <- sd(ess_trust_parl$trstprl)
```

n = 1727

Sample mean = 4.15

Sample SD = 2.485

Example: Trust in parliament (cont.)



The sample mean ($\bar{x} = 4.15$) looks close to 5 — but is it significantly different?

Example: Trust in parliament (cont.)

Step 1: State hypotheses:

- $H_0 : \mu = 5$ (average trust equals 5)
- $H_A : \mu \neq 5$ (average trust differs from 5)

Step 2: Choose α : 0.05

Step 3: Calculate test statistic $\frac{\bar{x} - \mu_0}{s/\sqrt{n}}$:

```
1 mu0 <- 5
2 t_stat <- (xbar - mu0) / (s / sqrt(n))
```

Test statistic: $t = -14.218$

Step 4: Calculate p-value:

```
1 # Two-sided p-value
2 p_value <- 2 * pt(-abs(t_stat), df = n - 1)
```

P-value = $<2e-16$

Example: Trust in parliament (cont.)

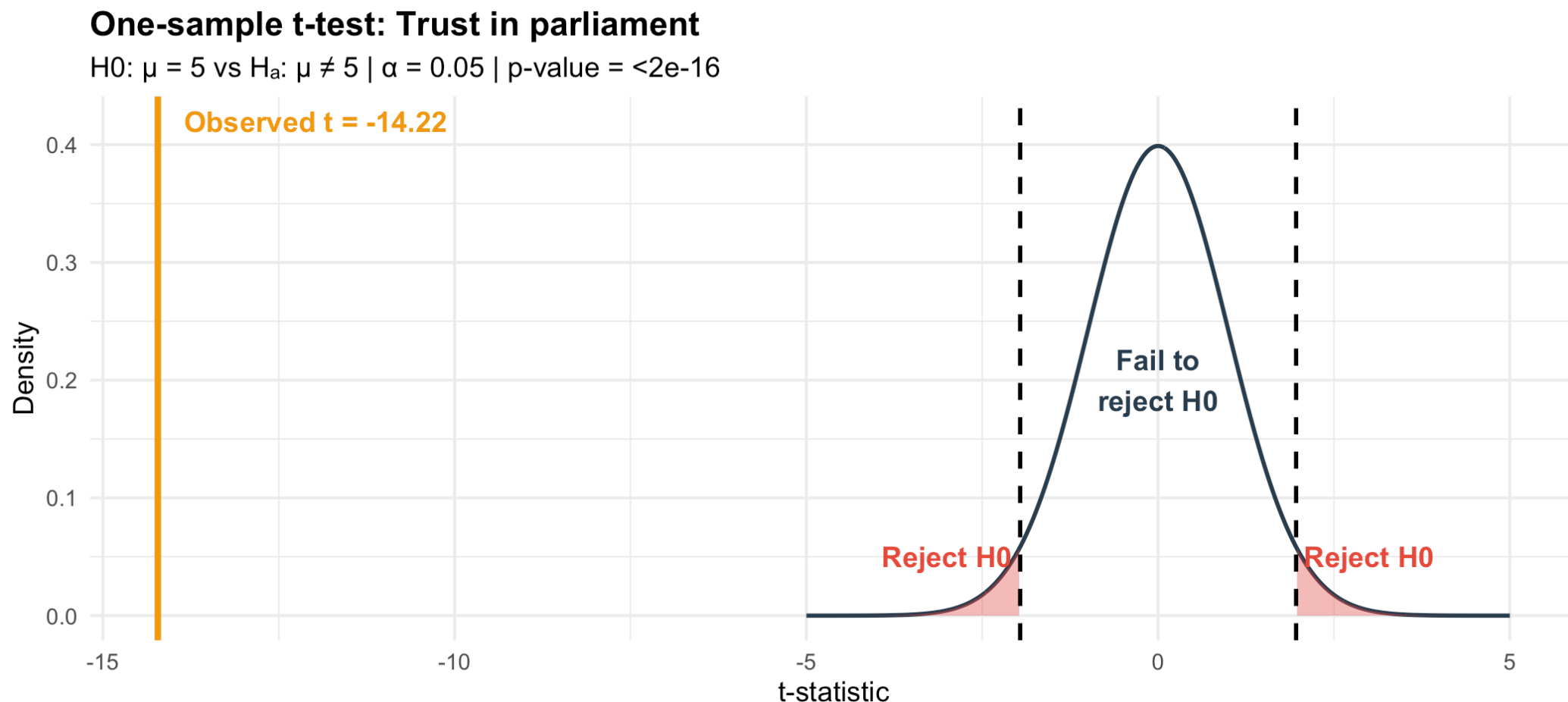
Using R's built-in function:

```
1 # Perform one-sample t-test  
2 t.test(ess_trust_parl$trstprl, mu = 5)
```

One Sample t-test

```
data:  ess_trust_parl$trstprl  
t = -14.218, df = 1726, p-value < 2.2e-16  
alternative hypothesis: true mean is not equal to 5  
95 percent confidence interval:  
 4.032712 4.267230  
sample estimates:  
mean of x  
 4.149971
```

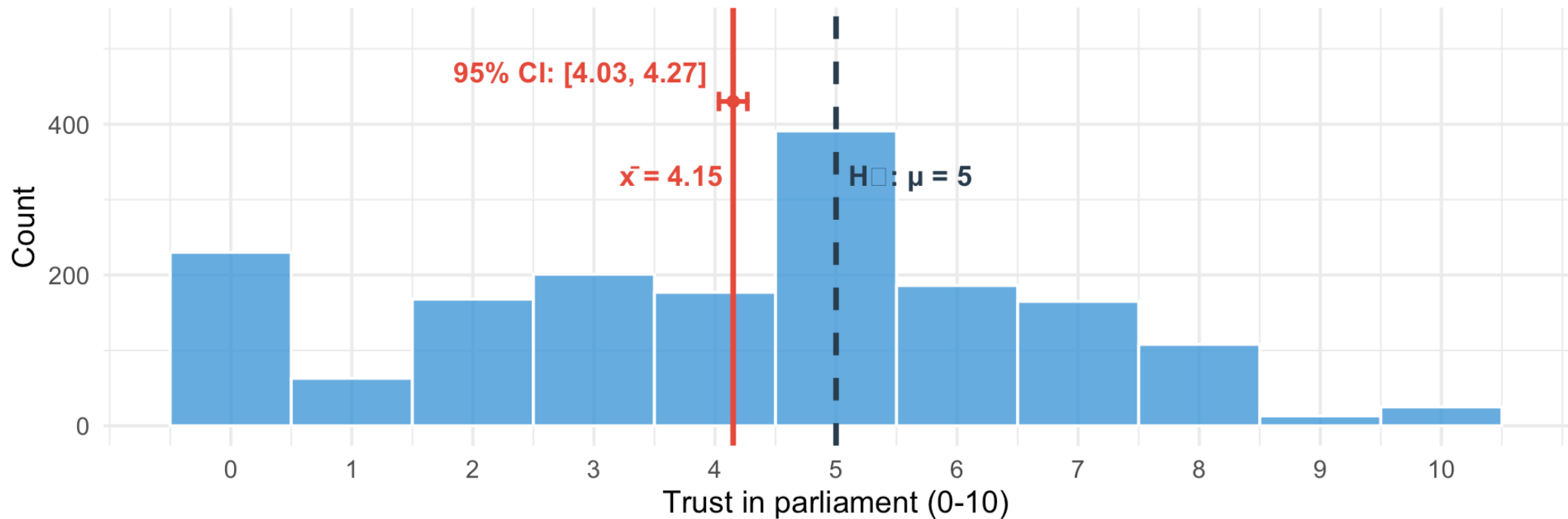
Visualizing the result



Decision: Reject H_0 ($p < 0.05$) at the 5% level. The data provide strong evidence that average trust in parliament differs from 5. Can we reject at the 1% level?

CI and hypothesis testing

Distribution of trust in parliament with 95% CI
European Social Survey Round 11 (France)



💡 The CI-hypothesis test duality

The 95% CI **does not contain** $\mu_0 = 5 \Rightarrow$ we **reject** H_0 at $\alpha = 0.05$.

More generally: a two-sided test at level α rejects H_0 **if and only if** μ_0 falls outside the $(1 - \alpha)$ CI.

Quick example: Student sleep

Health guidelines recommend at least **7 hours** of sleep per night. You survey a random sample of **40 students** and find that they report sleeping an average of **6 hours and 18 minutes** per night, with a standard deviation of **1 hour and 24 minutes**. Test at $\alpha = 0.05$ whether students sleep less than the recommended amount.

Step 1: State the hypothesis

$$H_0 : \mu = 7 \text{ vs } H_A : \mu < 7 \text{ (one-sided)}$$

Step 2: Choose significance level

$$\alpha = 0.05$$

Step 3: Compute the test statistic:

$$t = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}} = \frac{6.3 - 7}{1.4 / \sqrt{40}} \approx -3.16$$

Step 4: Compute the p-value (one-sided, left tail):

```
1 pt(t_sl, df = n_sl - 1)
[1] 0.001513711
```

Steps 5–6: Make a decision $\rightarrow$ reject or fail to reject H_0 ?

$p \approx 0.002 < 0.05 \rightarrow$ reject H_0 . Strong evidence that students sleep significantly less than the recommended 7 hours.

One-sample test for proportions

Goal: Test whether a population proportion p equals a specific hypothesized value p_0 .

Hypotheses:

$$H_0 : p = p_0 \quad \text{vs} \quad H_A : p \neq p_0$$

Test statistic (large sample):

$$z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1 - p_0)}{n}}}$$

where $\hat{p}$ is the sample proportion. (Denominator uses p_0 because H_0 assumed to be true.)

Under H_0 : $z \sim N(0, 1)$ approximately (when $np_0 \geq 10$ and $n(1 - p_0) \geq 10$)

Example: High interpersonal trust

Research question: Is the proportion of French adults with high interpersonal trust greater than 20%?

```
1 # Create high trust indicator
2 ess_trust <- ess_france |>
3   filter(pltrst %in% 0:10) |>
4   mutate(high_trust = pltrst >= 7) |>
5   select(high_trust)
6
7 # Summary
8 n_trust <- nrow(ess_trust)
9 p_hat <- mean(ess_trust$high_trust)
10 num_high_trust <- sum(ess_trust$high_trust)
```

n = 1760

Number with high trust = 363

Sample proportion = 0.206

Example: High interpersonal trust (cont.)

Testing $H_0 : p \leq 0.20$ vs $H_A : p > 0.20$.

Manual calculation first:

```
1 p0 <- 0.20
2
3 # Step 3: Test statistic
4 z_stat <- (p_hat - p0) / sqrt(p0 * (1 - p0) / n_trust)
```

z = 0.656

```
1 # Step 4: One-sided p-value
2 p_value <- 1 - pnorm(z_stat)
```

p-value = 0.256

Confirm with **prop.test()**:

```
1 prop.test(x = num_high_trust, n = n_trust, p = 0.20, alternative = "greater", correct = FALSE)
```

1-sample proportions test without continuity correction

data: num_high_trust out of n_trust, null probability 0.2

X-squared = 0.42969, df = 1, p-value = 0.2561

alternative hypothesis: true p is greater than 0.2

95 percent confidence interval:

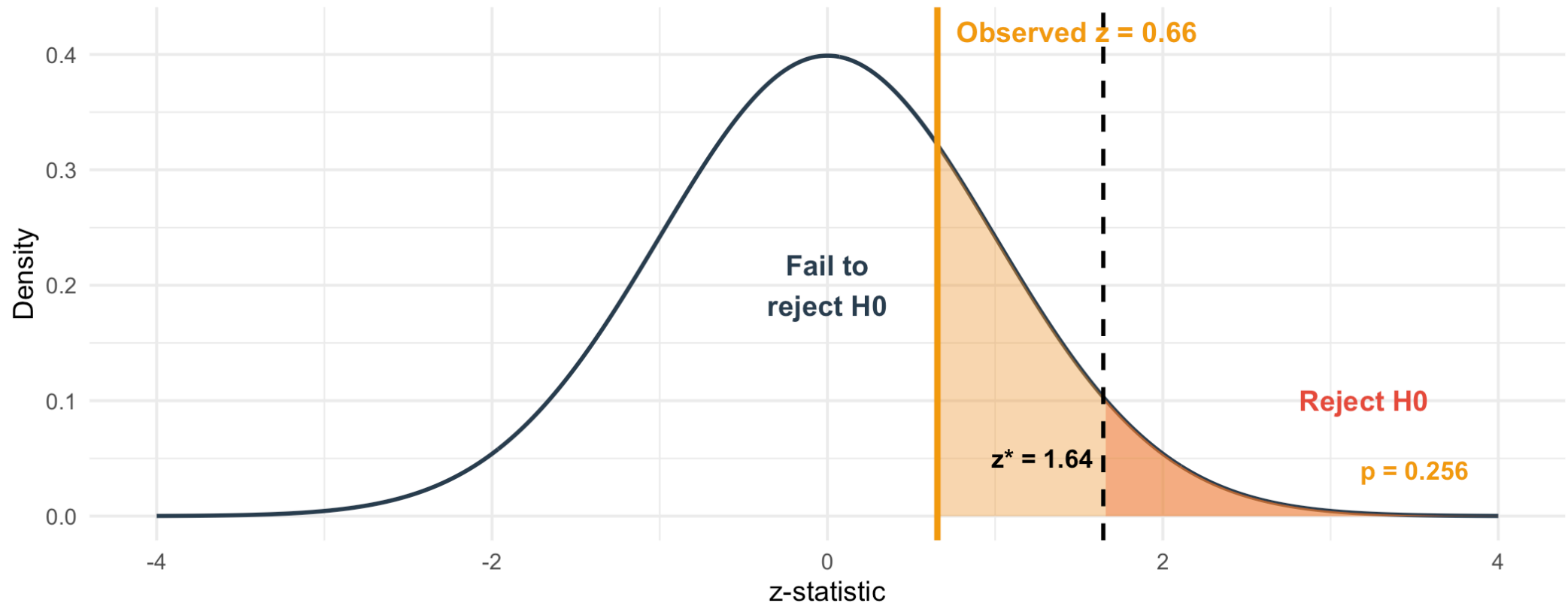
0.1908427 1.0000000

sample estimates:

Visualizing the result

One-sample z-test: High interpersonal trust

$H_0: p = 0.20$ vs $H_A: p > 0.20$ | $\alpha = 0.05$



Decision: Fail to reject H_0 ($p > 0.05$) at 5% level. No strong evidence that the proportion with high interpersonal trust exceeds 20%.

Your turn! #2

15:00

1. Load the ESS France data and `tidyverse`. Select the life satisfaction variable (`stflife`) keeping only values between 0 and 10. The goal is to test whether average life satisfaction equals 6.5 (use $\alpha = 0.05$).

```
1 library(tidyverse)
2
3 ess_france <- read.csv("https://www.dropbox.com/scl/fi/25h4lezq3zuzla94ejmqq/ess_france_11ed.csv?rlke
```

1. State the null and alternative hypotheses.
2. **Manually compute** the test statistic $t = (\bar{x} - \mu_0) / (s / \sqrt{n})$ and the two-sided p-value using `pt()`.
3. Verify your result with `t.test()`.
4. Make a decision and interpret.
5. Compute the 95% confidence interval for overage life satisfaction. Compare it with the decision you made above.

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Two-sample tests

Two-sample t-test: Independent groups

Goal: Test whether two population means are equal: $\mu_1 = \mu_2$

Hypotheses:

$$H_0 : \mu_1 = \mu_2 \quad \text{vs} \quad H_A : \mu_1 \neq \mu_2$$

Or equivalently: $H_0 : \mu_1 - \mu_2 = 0$ vs $H_A : \mu_1 - \mu_2 \neq 0$

Test statistic:

$$T = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{SE(\bar{X}_1 - \bar{X}_2)} = \frac{(\bar{X}_1 - \bar{X}_2) - 0}{SE(\bar{X}_1 - \bar{X}_2)}$$

Two versions:

1. **Equal variances** (pooled t-test): Assumes $\sigma_1^2 = \sigma_2^2$
2. **Unequal variances** (Welch's t-test): Does not assume equal variances (default in R)

Standard error for difference in means

Unequal variances – Welch's t-test (recommended default in R):

$$SE_{\text{Welch}} = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

Degrees of freedom are approximated by the Welch-Satterthwaite equation we discussed in the confidence interval lecture (R handles this automatically).

Equal variances – pooled t-test (only if variances are truly equal):

$$SE_{\text{pooled}} = s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}, \quad s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}$$



Tip

Rule of thumb: Always use Welch's t-test. It reduces to the pooled test when variances are equal, so there is no downside to using it as the default. In R, `t.test()` uses Welch's test by default.

Example: Education and trust

Research question: Does interpersonal trust differ between those with and without higher education?

```

1 # Create education groups
2 ess_educ_trust <- ess_france |>
3   filter(ppltrst %in% 0:10) |>
4   mutate(higher_educ = case_when(education == "High school diploma or less" ~ "No higher education",
5                                   education != "High school diploma or less" ~ "Higher education")) |>
6   filter(!is.na(higher_educ)) |>
7   select(ppltrst, higher_educ)
8
9 # Summary statistics by group
10 ess_educ_trust |>
11   group_by(higher_educ) |>
12   summarise(n = n(),
13             mean = mean(ppltrst),
14             sd = sd(ppltrst),
15             se = sd / sqrt(n))

```

A tibble: 2 × 5

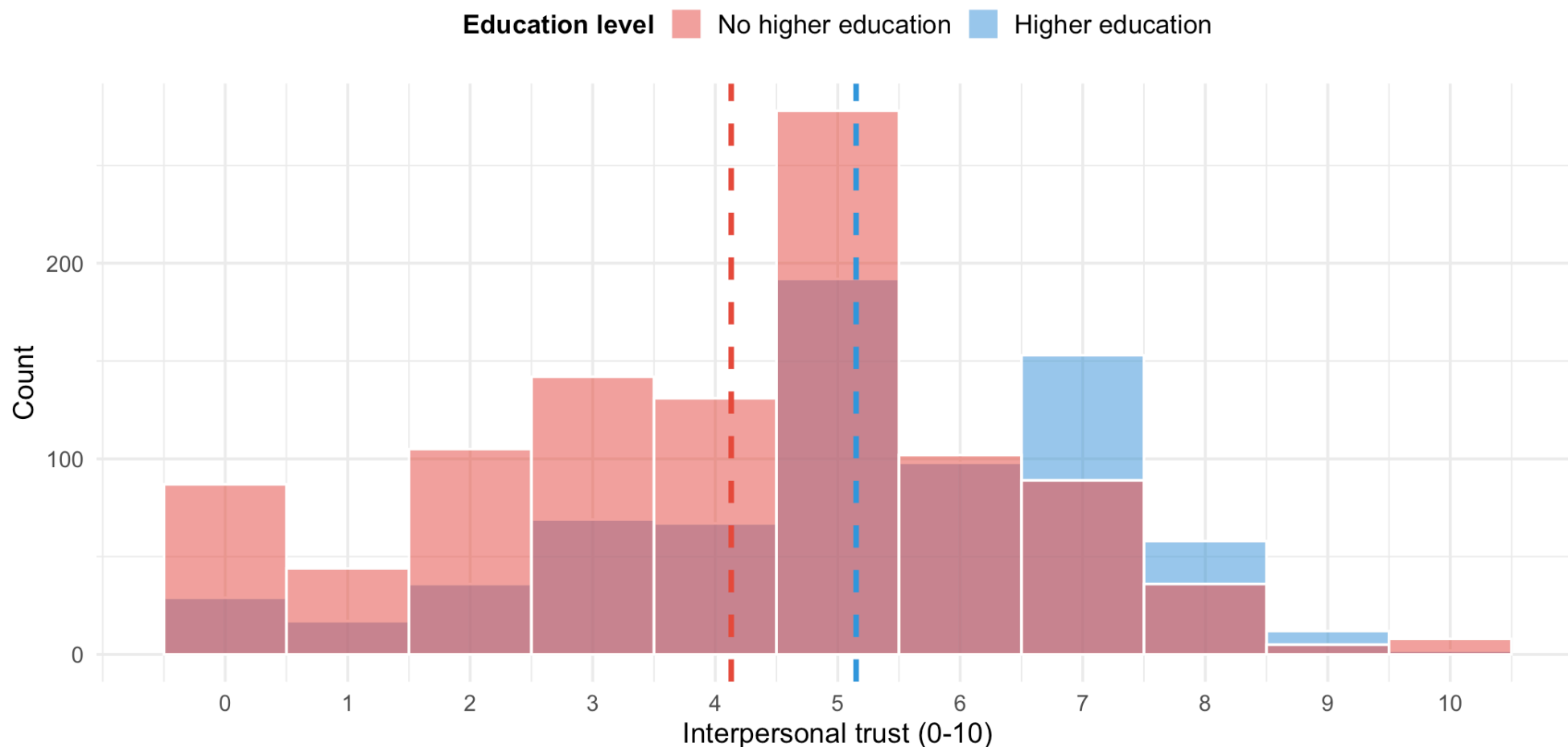
higher_educ	n	mean	sd	se
<chr>	<int>	<dbl>	<dbl>	<dbl>
1 Higher education	733	5.15	2.07	0.0763
2 No higher education	1027	4.13	2.16	0.0676

Example: Education and trust (cont.)

Visualize the distributions — do the groups look different?

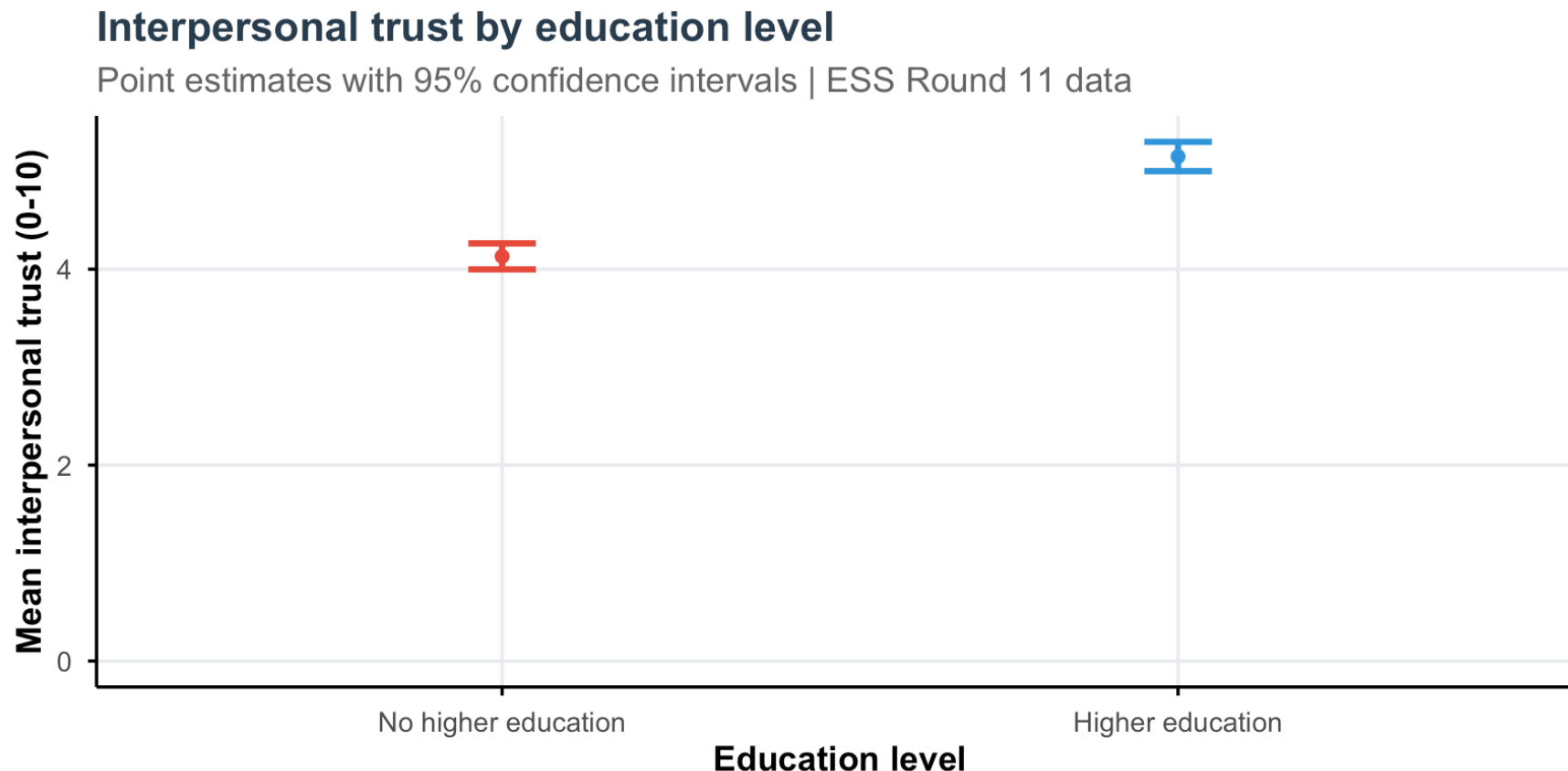
Distribution of interpersonal trust by education level

European Social Survey Round 11 (France)



Example: Education and trust (cont.)

Before testing — visualize the group means with 95% CIs:



The CIs don't overlap — strong visual evidence of a difference. Let's confirm with a formal test.

Example: Education and trust (cont.)

Manual computation – Welch's t-test: Confirm with `t.test()`:

```
1 # Group statistics
2 stats <- educ_summary
3
4 n1 <- stats$n[1]
5 xbar1 <- stats$mean[1]
6 s1 <- stats$sd[1]
7 n2 <- stats$n[2]
8 xbar2 <- stats$mean[2]
9 s2 <- stats$sd[2]
10
11 # SE (Welch)
12 se <- sqrt(s1^2/n1 + s2^2/n2)
```

SE = 0.1019

```
1 # Test statistic
2 t_stat <- (xbar1 - xbar2) / se
```

t = 10.007

```
1 # Approximate p-value (using large-sample z app
2 p_value <- 2 * pnorm(-abs(t_stat))
```

p-value ≈ <2e-16

(uses exact Welch-Satterthwaite df)

```
1 t.test(ppltrst ~ higher_educ,
2       data = ess_educ_trust)
```

Welch Two Sample t-test

data: ppltrst by higher_educ

t = 10.007, df = 1619.7, p-value < 2.2e-16

alternative hypothesis: true difference in means
between group Higher education and group No higher
education is not equal to 0

95 percent confidence interval:

0.8197456 1.2194366

sample estimates:

mean in group Higher education mean in group No
higher education

5.150068

4.130477

Conclusion: Those with higher education have significantly higher trust (difference ≈ 0.67, $p < 0.001$).

Quick example: Study hours and exam outcome

A university records the weekly study hours of students who sat a statistics exam. Among the **40 students who passed**, the average was **5.3 hours** per day with a standard deviation of **1.8 hours**. Among the **30 students who failed**, the average was **3.2 hours** per day with a standard deviation of **1.5 hours**. Test at $\alpha = 0.05$ whether students who passed studied more on average.

Step 1: State the hypotheses:

$$H_0 : \mu_1 = \mu_2 \text{ vs } H_A : \mu_1 > \mu_2 \text{ (one-sided)}$$

Step 2: Choose significance level:

$$\alpha = 0.05$$

Step 3: Compute the test statistic:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{SE} = \frac{5.3 - 3.2}{\sqrt{\frac{1.8^2}{40} + \frac{1.5^2}{30}}} \approx 5.32$$

Step 4: Compute the p-value (one-sided, right tail; use $df = 67.19$ (Welch-Satterthwaite)):

```
1 1 - pt(t_ts, df = 67.19)
```

```
[1] 6.491827e-07
```

Steps 5–6: Make a decision $\rightarrow$ reject or fail to reject H_0 ?

$p < 0.001 < 0.05 \rightarrow$ reject H_0 . Strong evidence that students who passed studied significantly more hours per day than those who failed.

Two-sample test for proportions

Goal: Test whether two population proportions are equal: $p_1 = p_2$

Test statistic:

$$z = \frac{\hat{p}_1 - \hat{p}_2}{SE}, \quad SE = \sqrt{\hat{p}(1 - \hat{p}) \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}$$

where $\hat{p} = \frac{x_1 + x_2}{n_1 + n_2}$ is the **pooled proportion** (estimated under $H_0 : p_1 = p_2$).

Under H_0 : $z \sim N(0, 1)$ approximately (requires $n_i \hat{p}(1 - \hat{p}) \geq 5$ for each group)

Example: High trust by education

Research question: Does the proportion with high interpersonal trust differ by education?

```

1 # Create high trust indicator by education
2 ess_trust_educ <- ess_educ_trust |>
3   mutate(high_trust = ppltrst >= 7)
4
5 # Summary
6 prop_table <- ess_trust_educ |>
7   group_by(higher_educ) |>
8   summarise(n = n(),
9             n_high_trust = sum(high_trust),
10            prop_high_trust = mean(high_trust))
11
12 prop_table

```

```

# A tibble: 2 × 4
  higher_educ      n n_high_trust prop_high_trust
  <chr>      <int>      <int>      <dbl>
1 Higher education    733         225      0.307
2 No higher education 1027         138      0.134

```

Example: High trust by education (cont.)

Manual calculation first:

```

1 # Group proportions
2 # Higher education
3 p1_hat <- prop_table$prop_high_trust[1]
4 # No higher education
5 p2_hat <- prop_table$prop_high_trust[2]
6
7 n_high1 <- prop_table$n_high_trust[1]
8 n_high2 <- prop_table$n_high_trust[2]
9 n1 <- prop_table$n[1]
10 n2 <- prop_table$n[2]
11
12 # Pooled proportion (under H0: p1 = p2)
13 p_pool <- (n_high1 + n_high2) / (n1 + n2)

```

Pooled proportion: 0.206

```

1 # SE and z-statistic
2 SE_prop <- sqrt(p_pool * (1 - p_pool) * (1/n1 +
3 z_stat_prop <- (p1_hat - p2_hat) / SE_prop

```

z = 8.822

```
1 p_value <- 2 * pnorm(-abs(z_stat_prop))
```

p-value = <2e-16

Verify with `prop.test()`:

```

1 prop.test(x = prop_table$n_high_trust,
2           n = prop_table$n,
3           correct = FALSE)

```

2-sample test for equality of proportions
without continuity correction

data: prop_table\$n_high_trust out of prop_table\$n
X-squared = 77.82, df = 1, p-value < 2.2e-16

alternative hypothesis: two.sided

95 percent confidence interval:

0.1332162 0.2119553

sample estimates:

prop 1	prop 2
0.3069577	0.1343720

Conclusion: The proportion with high trust is significantly higher among those with higher education ($p < 0.001$).

Your turn! #3

15:00

Part 1: Test whether average life satisfaction differs between men and women:

1. Calculate summary statistics by gender
2. Compute a 95% confidence interval for both means. Do they overlap?
3. Perform a two-sample t-test manually (using $df = 1730.8$) and using `t.test`.
4. Interpret: is the difference statistically significant? At what level? Practically meaningful?

Part 2: Test whether the proportion who are “very satisfied” with democracy ($stf_{dem} \geq 8$) differs between those with and without higher education:

1. Compute 95% confidence intervals for both proportions. Do they overlap?
2. Perform a two-sample proportion test manually and using `prop.test`.
3. Interpret: is the difference statistically significant? At what level? Practically meaningful?

Today's lecture

The big question: How do we test whether our data provides evidence for or against a claim about the population?

1. Introduction to hypothesis testing
2. The hypothesis testing framework
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Paired tests

Paired vs independent samples

Independent samples: Two separate groups with no relationship between observations (e.g., men vs women)

Paired samples: Two measurements on the same individuals or matched pairs (e.g., before vs after, parliament vs politicians)

Why does it matter?

- Paired data has **less variability** because we control for individual differences
- Use a different test that accounts for the pairing
- More **statistical power** to detect differences

! Key principle

With paired data, we analyze the **differences within pairs**, not the two groups separately.

Paired t-test: Theory

Goal: Test whether the mean difference between paired observations equals a hypothesized value δ_0 .

Hypotheses:

$$H_0 : \mu_d = \delta_0 \quad \text{vs} \quad H_A : \mu_d \neq \delta_0$$

where $\mu_d = \mu_1 - \mu_2$ is the population mean of the pairwise differences.

Typically, $\delta_0 = 0$ (no difference).

Method:

1. Calculate differences: $d_i = x_{1i} - x_{2i}$ for each pair i
2. Perform a one-sample t-test on the differences

Test statistic: $t = \frac{\bar{d} - \delta_0}{s_d / \sqrt{n}}$

When $\delta_0 = 0$ this simplifies to $t = \frac{\bar{d}}{s_d / \sqrt{n}}$.

Example: Trust in parliament vs politicians

Research question: Do French adults trust parliament and politicians equally?

```
1 # Prepare paired data
2 ess_paired <- ess_france |>
3   filter(trstprl %in% 1:10 & trstplt %in% 0:10) |>
4   select(trstprl, trstplt) |>
5   mutate(diff = trstprl - trstplt)
6
7 # Summary statistics
```

n = 1490

Mean trust in parliament: 4.783

Mean trust in politicians: 4.128

Mean difference: 0.655

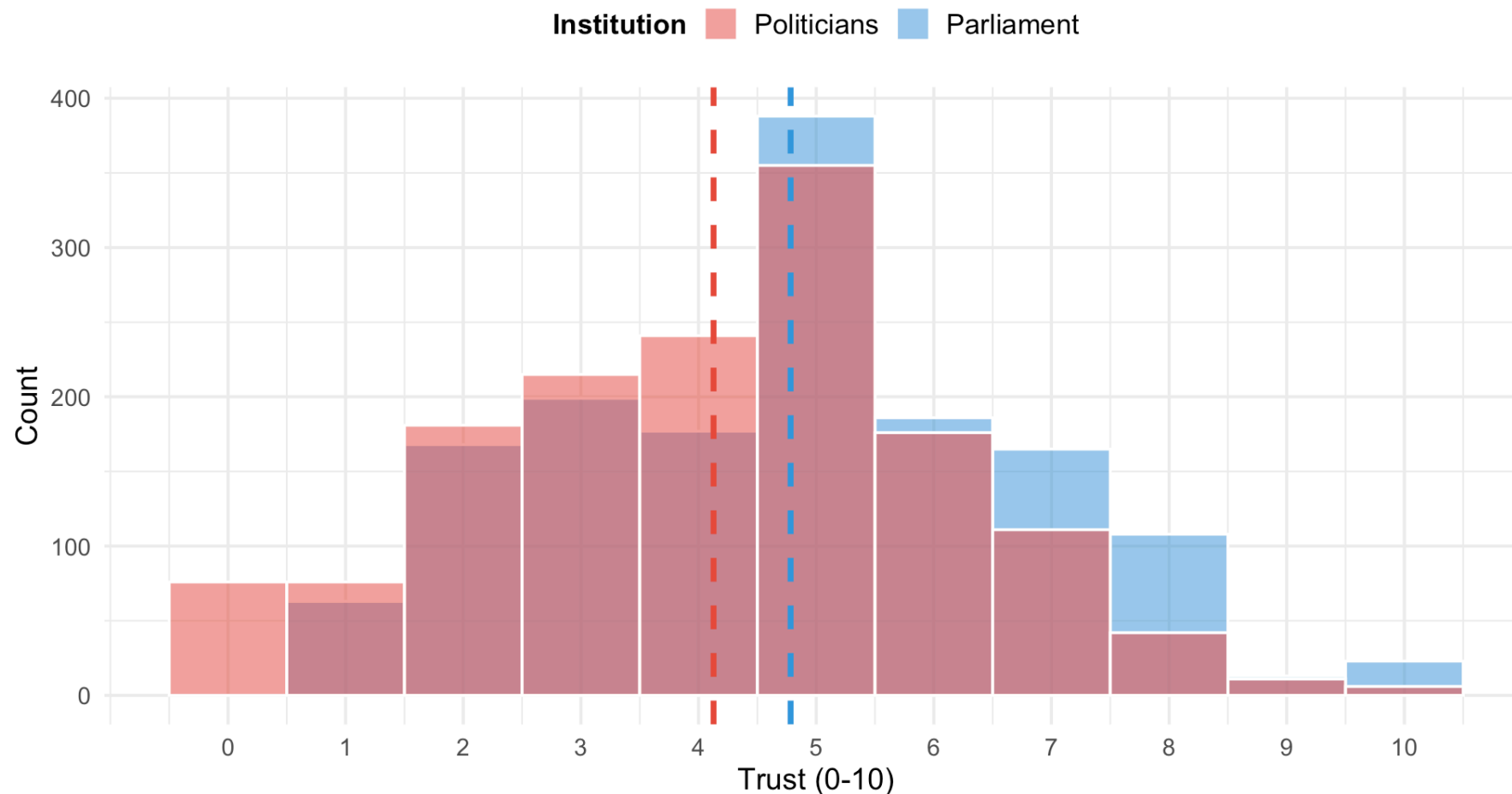
SD of differences: 1.892

Example: Trust comparison (cont.)

Visualize the paired data:

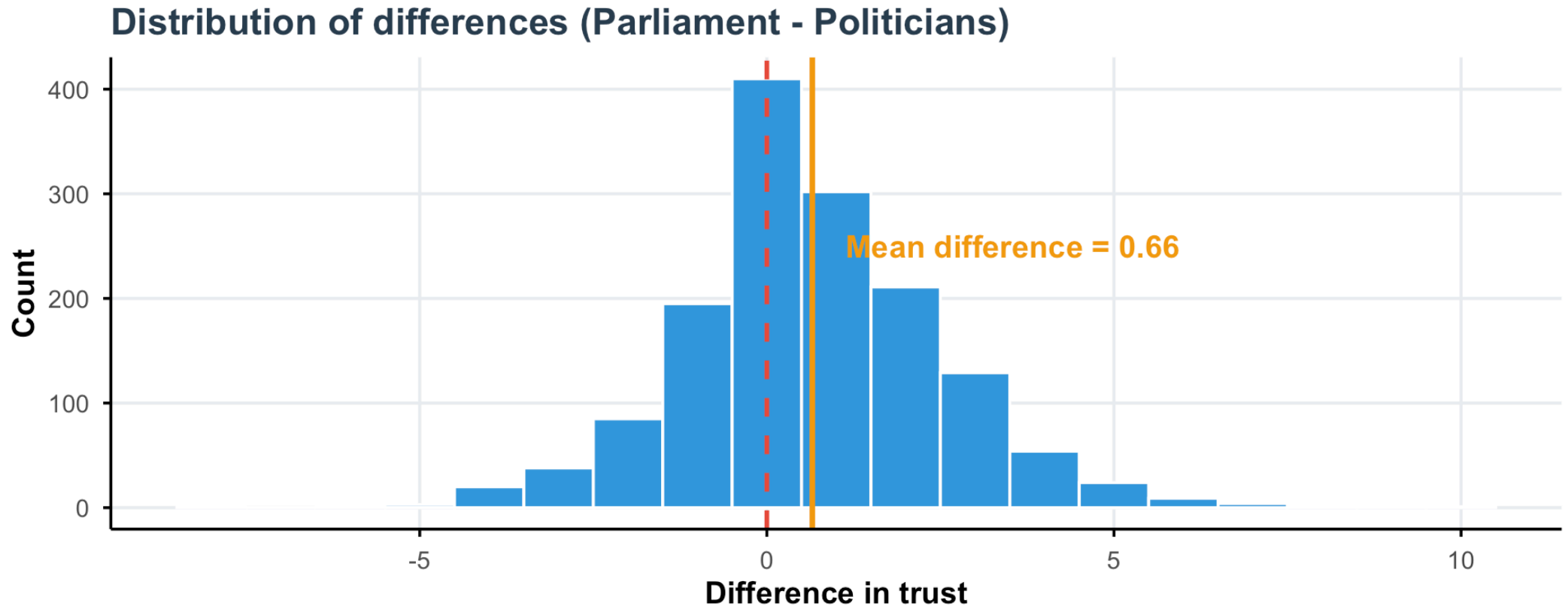
Distribution of trust in parliament vs politicians

European Social Survey Round 11 (France)



Example: Trust comparison (cont.)

Visualize the differences:



Example: Trust comparison (cont.)

Paired t-test = one-sample t-test on the differences $D_i = \text{Parliament} - \text{Politicians}$

Parliament: $\bar{x} = 4.78$ **Politicians:** $\bar{x} = 4.13$

$$\bar{d} = 4.78 - 4.13 = 0.66, \quad s_d = 1.89, \quad n = 1490$$

$$t = \frac{\bar{d}}{s_d / \sqrt{n}} = \frac{0.655}{1.892 / \sqrt{1490}} \approx 13.36, \quad p < 0.001$$

Your turn! #4

10:00

Test whether satisfaction with the state of the economy (`stfeco`) differs from satisfaction with the government (`stfgov`).

1. Create a new variable equal to the difference between the two satisfaction measures. Make sure to drop values not between 0 and 10.
2. Perform a paired t-test manually and using `t.test`.
3. Interpret: Are people more satisfied with the economy or government?

Today's lecture

The big question: How do we test whether our data provides evidence for or against a claim about the population?

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Chi-square test of independence

Testing relationships between categorical variables

So far: Testing hypotheses about means and proportions

New question: Are two categorical variables **independent**, or is there an **association** between them?

Examples:

- Is trust level (low/medium/high) independent of education (lower/higher)?
- Is life satisfaction (dissatisfied/neutral/satisfied) independent of gender?
- Is voting intention independent of age group?

Tool: **Chi-square test of independence** (also called chi-square test of association)

Contingency tables

A **contingency table** (or cross-tabulation) shows the frequency distribution of two categorical variables.

Example: Trust level by education

```
1 # Create trust categories
2 ess_trust_cat <- ess_educ_trust |>
3   filter(ppltrst %in% 0:10) |>
4   mutate(trust_level = case_when(
5     ppltrst <= 3 ~ "Low trust (0-3)",
6     ppltrst <= 6 ~ "Medium trust (4-6)",
7     ppltrst <= 10 ~ "High trust (7-10)"
8   ))
9
10 # Create contingency table
11 cont_table <- table(ess_trust_cat$higher_educ, ess_trust_cat$trust_level)
12 cont_table
```

	High trust (7-10)	Low trust (0-3)	Medium trust (4-6)
Higher education	225	151	357
No higher education	138	378	511

Observed vs expected frequencies

Under independence: The expected frequency in each cell is:

$$E_{ij} = \frac{(\text{row } i \text{ total}) \times (\text{column } j \text{ total})}{\text{grand total}}$$

Intuition: Independence means $P(\text{row } i \cap \text{col } j) = P(\text{row } i) \times P(\text{col } j)$.

Estimating marginal probabilities from the data:

$$E_{ij} = \underbrace{N}_{\# \text{ obs.}} \times \underbrace{\frac{R_i}{N}}_{\hat{P}(\text{row } i)} \times \underbrace{\frac{C_j}{N}}_{\hat{P}(\text{col } j)} = \frac{R_i \times C_j}{N}$$

i.e. “how many observations *should* be in this cell if the two variables were unrelated?”

Expected frequencies under independence

```
1 # Add marginal totals
2 addmargins(cont_table)
```

	High trust (7-10)	Low trust (0-3)	Medium trust (4-6)	Sum
Higher education	225	151	357	733
No higher education	138	378	511	1027
Sum	363	529	868	1760

```
1 # Calculate expected frequencies
2 expected <- chisq.test(cont_table)$expected
3 round(expected, 1)
```

	High trust (7-10)	Low trust (0-3)	Medium trust (4-6)
Higher education	151.2	220.3	361.5
No higher education	211.8	308.7	506.5

Compare to observed:

```
1 cont_table
```

	High trust (7-10)	Low trust (0-3)	Medium trust (4-6)
Higher education	225	151	357
No higher education	138	378	511

Notice: More higher education people in “high trust” than expected!

Chi-square test statistic

Measures the discrepancy between observed and expected frequencies:

$$\chi^2 = \sum_{i,j} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

where:

- O_{ij} = observed frequency in cell (i, j)
- E_{ij} = expected frequency under independence

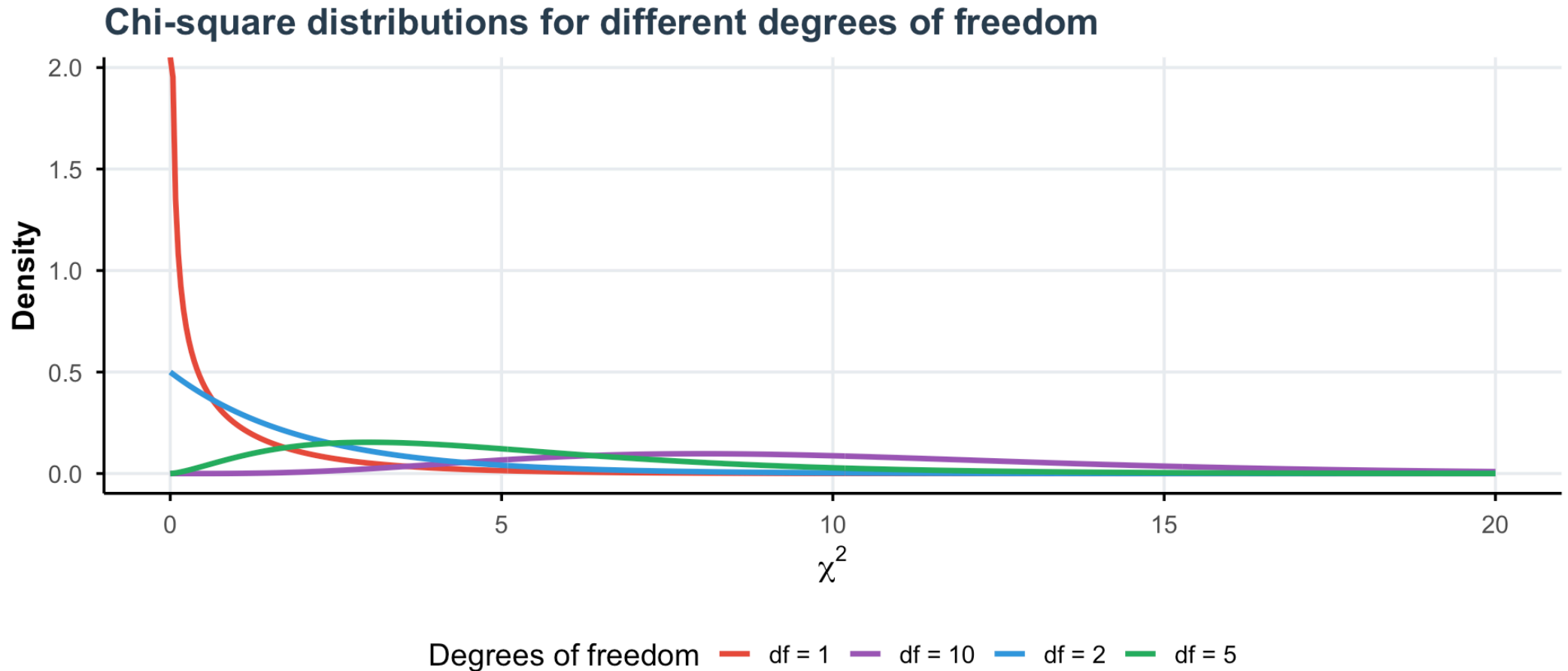
Under H_0 (independence):

$$\chi^2 \sim \chi^2_{(r-1)(c-1)}$$

where r = number of rows, c = number of columns.

Degrees of freedom: $(r - 1)(c - 1) = (2 - 1)(3 - 1) = 2$ for our example.

The chi-square distribution



Note: Chi-square distribution is right-skewed and only takes **positive values**.

Example: Trust by education (cont.)

Hypotheses:

- H_0 : Trust level and education are independent
- H_A : Trust level and education are associated

Manual computation:

```
1 # Observed and expected
2 O <- as.matrix(cont_table)
3 E <- chisq.test(cont_table)$expected
4
5 # Chi-square statistic: sum of (O - E)^2 / E
6 chi_manual <- sum((O - E)^2 / E)
```

$\chi^2 = 99.24$

```
1 df_chi <- (nrow(O) - 1) * (ncol(O) - 1)
2 p_value <- pchisq(chi_manual, df = df_chi, lower.tail = FALSE)
```

p-value = <2e-16

Confirm with `chisq.test()`:

```
1 chi_result <- chisq.test(cont_table)
2 chi_result
```

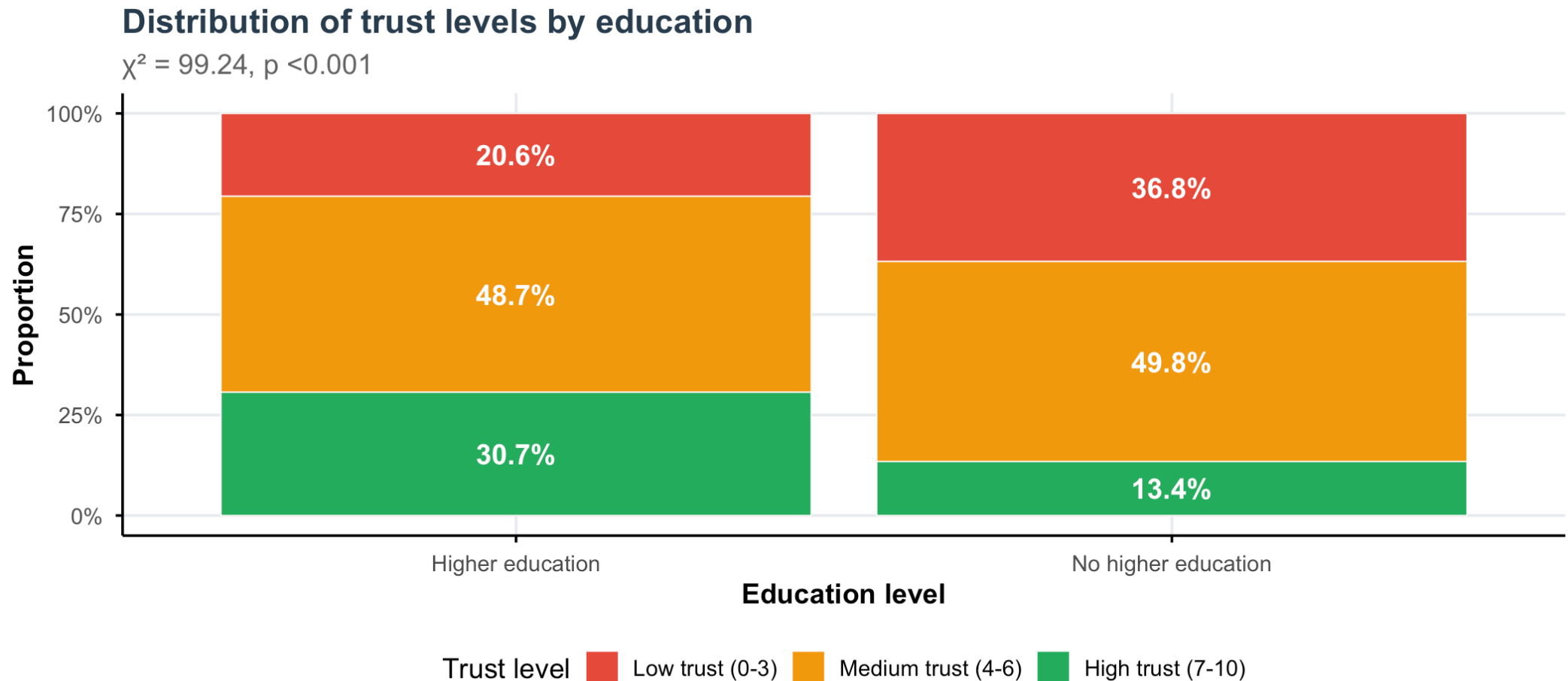
Pearson's Chi-squared test

data: cont_table

X-squared = 99.24, df = 2, p-value < 2.2e-16

We reject H_0 : trust level and education are **not independent** → people with higher education tend to report systematically different levels of trust in parliament.

Visualizing the association



Your turn! #5

15:00

Test whether life satisfaction is independent of gender:

1. Create a contingency table of life satisfaction categories (low: 0-4, medium: 5-7, high: 8-10) by gender.

```
1 ess_lifesat_cat <- ess_france |>
2   filter(stflife %in% 0:10, gndr %in% 1:2) |>
3   mutate(gender = recode(gndr, `1` = "Male", `2` = "Female"),
4           sat_level = case_when(stflife <= 4 ~ "Low (0-4)",
5                                 stflife <= 7 ~ "Medium (5-7)",
6                                 stflife <= 10 ~ "High (8-10)"))
```

2. Calculate expected frequencies by hand ($E_{ij} = \text{row total} \times \text{col total} / n$).
3. Compute the chi-square statistic manually using $\chi^2 = \sum (O - E)^2 / E$.
4. Find the p-value with `pchisq()`.
5. Verify with `chisq.test()`.
6. Interpret: Is there evidence of an association?

Today's lecture

The big question: How do we test whether our data provides evidence for or against a claim about the population?

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Randomization methods

Beyond theory-based tests

So far: We've used **theory-based tests** that rely on distributional assumptions (t-distribution, normal distribution, chi-square distribution)

Alternative approach: **Randomization tests** (also called permutation tests)

- Don't rely on distributional assumptions
- Use simulation to create the null distribution
- More intuitive: "What would we see if there were no effect?"

The **infer** package provides a tidy workflow for this approach!

The **infer** workflow

Every hypothesis test with **infer** follows **four steps**:

1. **specify()**: Specify the variables and success (for proportions)
2. **hypothesize()**: Declare the null hypothesis
3. **generate()**: Generate simulated samples under H_0
4. **calculate()**: Calculate the test statistic for each simulation

Then: **visualize()** and **get_p_value()**

Permutation test for two means

Logic: If education and trust are unrelated, we could randomly shuffle the education labels and get similar differences by chance.

Method:

1. Calculate observed difference in means
2. Randomly permute the group labels many times (e.g., 1,000)
3. Calculate the difference in means for each permutation
4. Compare observed difference to permutation distribution

P-value: Proportion of permutations with difference $\geq$ observed (in absolute value)

Example: Education and trust (permutation)

```

1 # Step 1: Calculate observed statistic
2 obs_diff <- ess_educ_trust |>
3   specify(ppltrst ~ higher_educ) |>
4   calculate(stat = "diff in means", order = c("Higher education", "No higher education"))
5
6 obs_diff

```

Response: ppltrst (numeric)

Explanatory: higher_educ (factor)

A tibble: 1 × 1

```

  stat
<dbl>
1  1.02

```

```

1 # Steps 2-4: Generate null distribution via permutation
2 set.seed(2024)
3 null_dist <- ess_educ_trust |>
4   specify(ppltrst ~ higher_educ) |>
5   hypothesize(null = "independence") |>
6   generate(reps = 10000, type = "permute") |>
7   calculate(stat = "diff in means", order = c("Higher education", "No higher education"))

```

Visualizing the null distribution

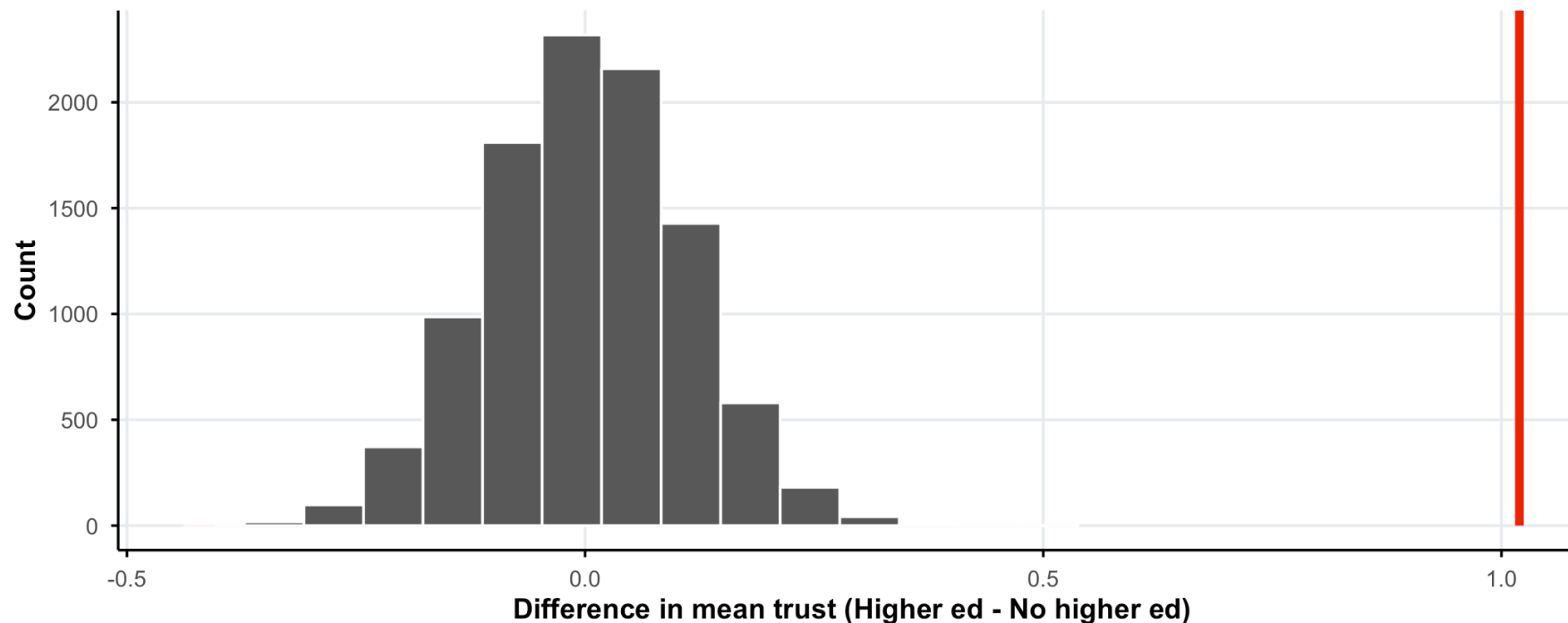
```

1 # Visualize
2 null_dist |>
3   visualize() +
4   shade_p_value(obs_stat = obs_diff, direction = "both") +
5   labs(title = "Permutation distribution of difference in means",
6         subtitle = "Red line shows observed difference",
7         x = "Difference in mean trust (Higher ed - No higher ed)",
8         y = "Count")

```

Permutation distribution of difference in means

Red line shows observed difference



Calculate p-value

```
1 # Get p-value
2 p_value_perm <- null_dist |>
3   get_p_value(obs_stat = obs_diff, direction = "both")
4
5 p_value_perm
```

A tibble: 1 × 1

	p_value <dbl>
1	0

Compare to theory-based test:

```
1 # Theory-based test (from earlier)
2 t.test(ppltrst ~ higher_educ, data = ess_educ_trust)$p.value
```

[1] 6.469825e-23

Very similar! Both show strong evidence of a difference ($p < 0.001$).

Theory-based vs randomization

When to use each approach?

Theory-based tests:

- Faster (no simulation needed)
- Well-established and widely recognized
- Require distributional assumptions
- Work well when assumptions are met

Randomization tests:

- More flexible (fewer assumptions)
- More intuitive interpretation
- Computationally intensive
- Ideal for small samples or non-normal data
- Exact p-values (not approximate)

In practice: Theory-based tests are usually fine if assumptions are met.

Use randomization when: sample is small, data is skewed, or you want to be conservative.

Your turn! #6

15:00

Perform a permutation test for the difference in life satisfaction between men and women using `infer`.

1. Calculate observed difference.
2. Generate 10,000 permutations
3. Visualize and calculate p-value.
4. Compare to theory-based t-test.

Today's lecture

The big question: How do we test whether our data provides evidence for or against a claim about the population?

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Statistical power and sample size

Statistical power

Statistical power = $1 - \beta$ = Probability of correctly rejecting a false null hypothesis

Factors affecting power:

1. **Effect size:** Larger true effects → easier to detect → higher power
2. **Sample size:** Larger samples → more information → higher power
3. **Significance level:** Larger α → easier to reject H_0 → higher power (but more Type I errors!)
4. **Variability:** Less noise → clearer signal → higher power

Rule of thumb:

Aim for power ≥ 0.80 (80% chance of detecting a true effect)

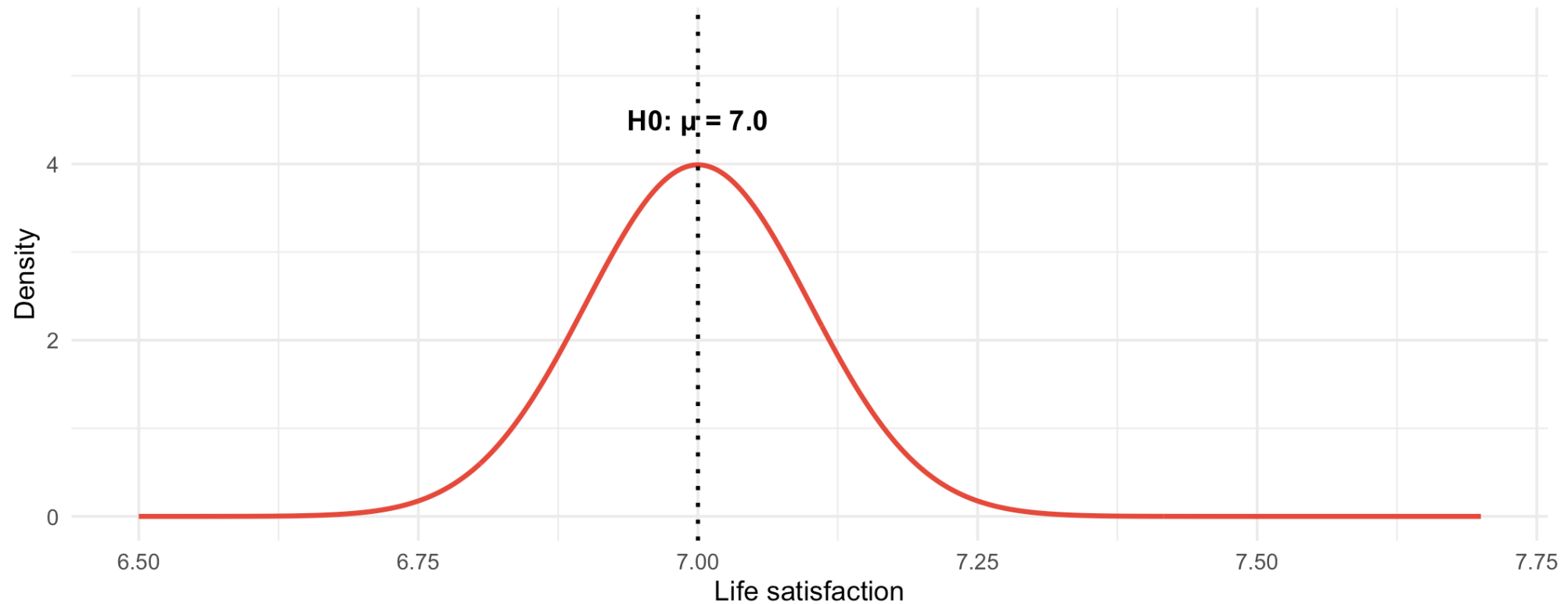
Power analysis visualization

Setup: Testing $H_0 : \mu = 7.0$ when the true value is $\mu = 7.3$ (effect size = 0.3, SE = 0.1)

Power analysis visualization (cont.)

Step 1: Null hypothesis distribution

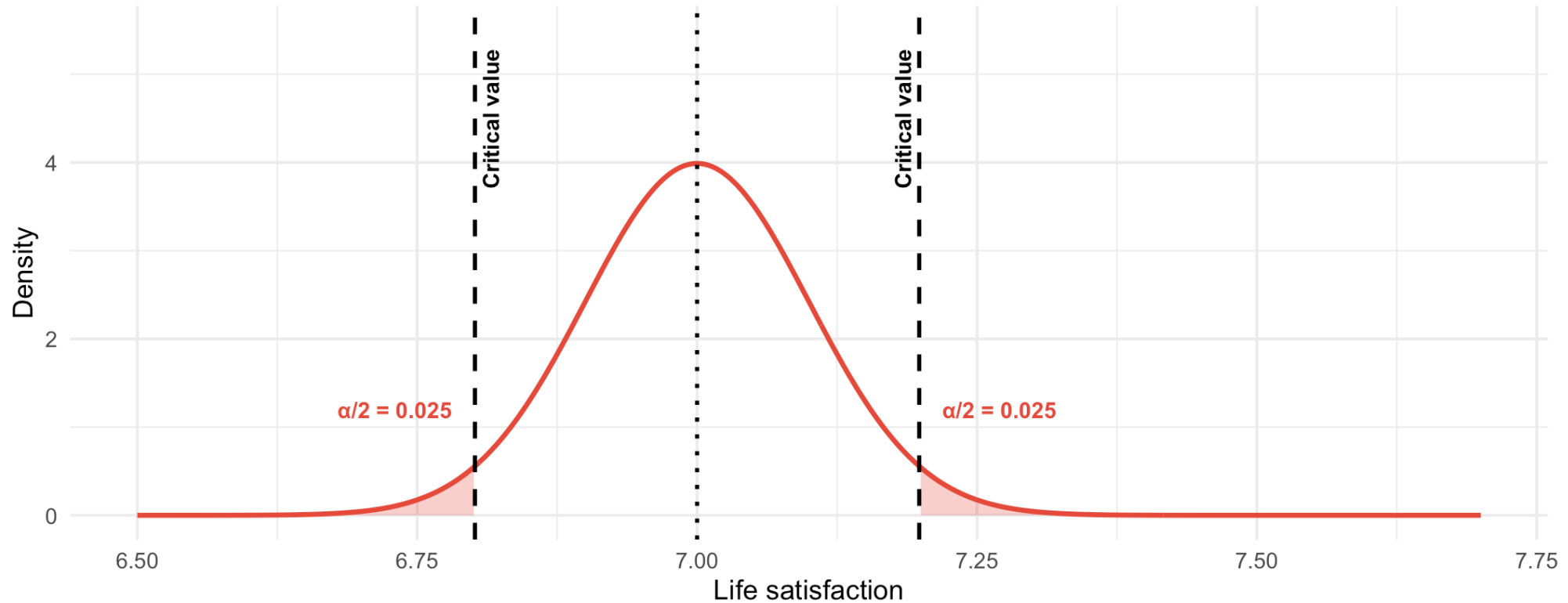
Centered around 7



Power analysis visualization (cont. 2)

Step 2: Add critical values and Type I error regions (two-sided test)

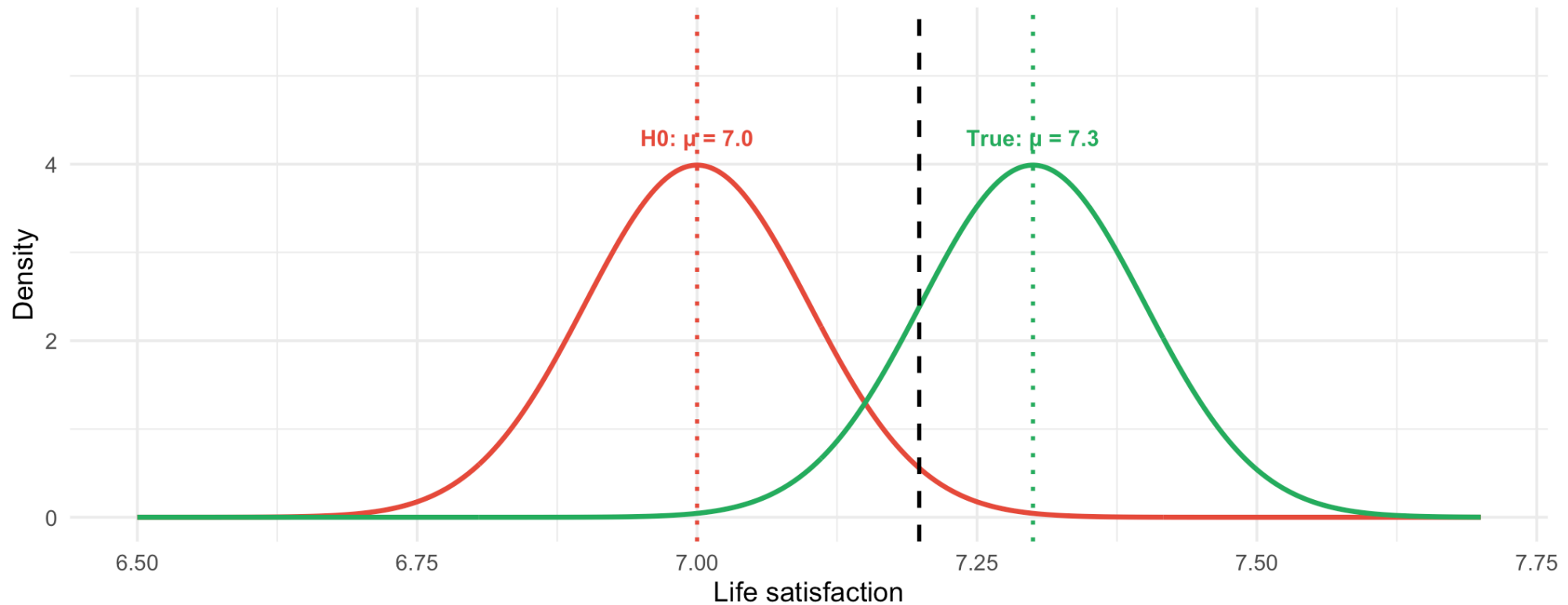
We reject H_0 if the sample mean falls in either tail — each has probability $\alpha/2 = 0.025$



Power analysis visualization (cont. 3)

Step 3: Add true distribution (if H0 is false)

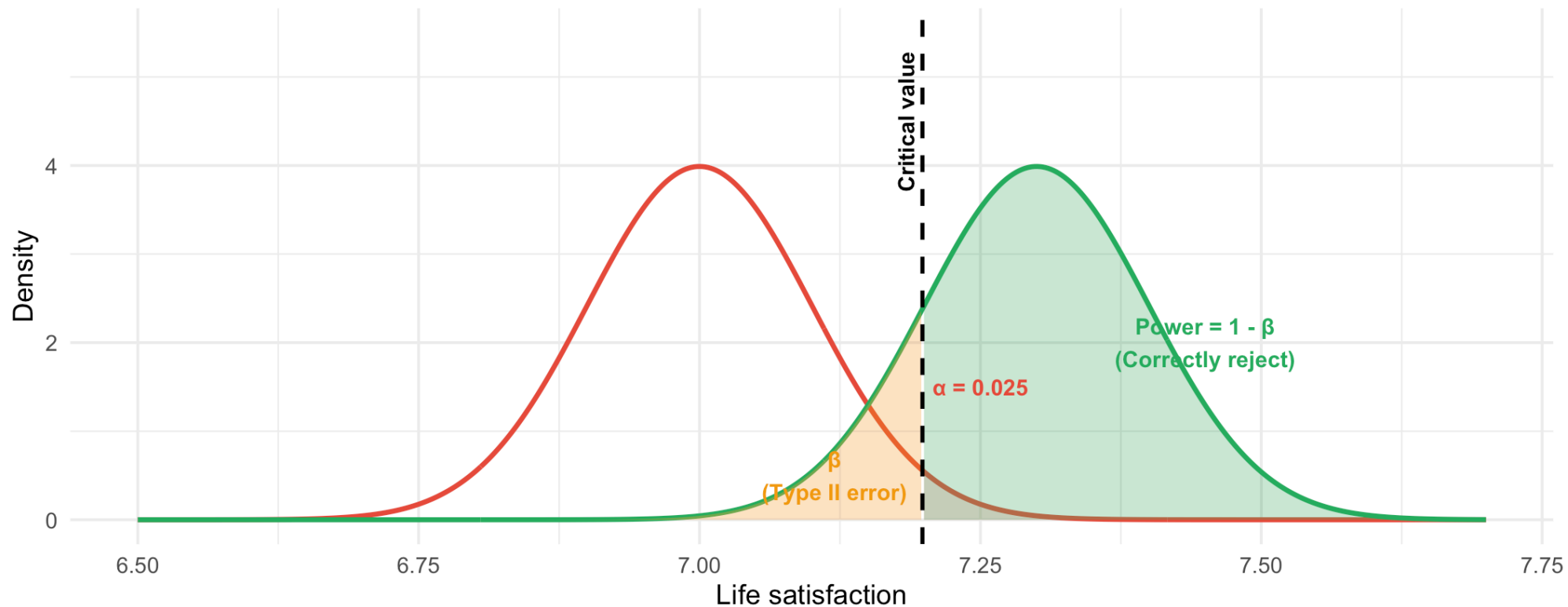
The true mean is actually 7.3, not 7.0



Power analysis visualization (cont. 4)

Step 4: Power and Type II error

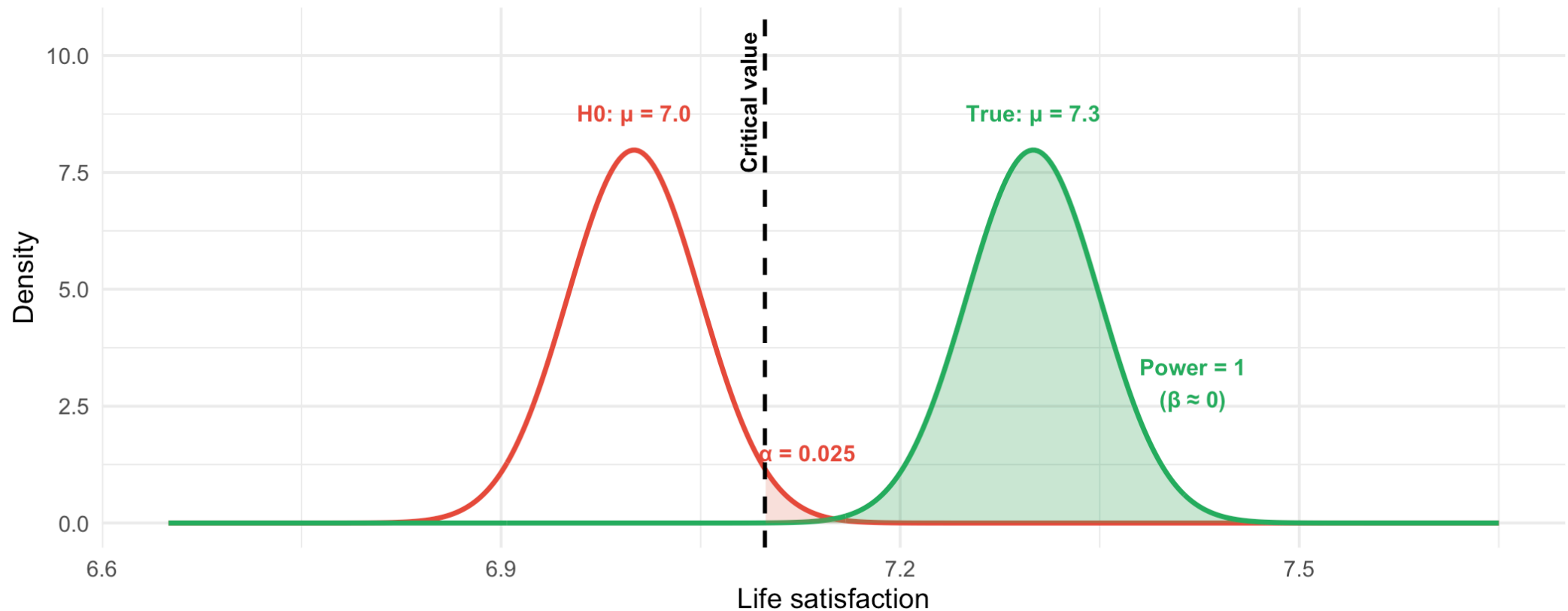
Power = probability of correctly rejecting when H_0 is false



Power analysis: effect of larger n

Larger $n \rightarrow$ smaller SE: distributions pull apart, power $\uparrow$

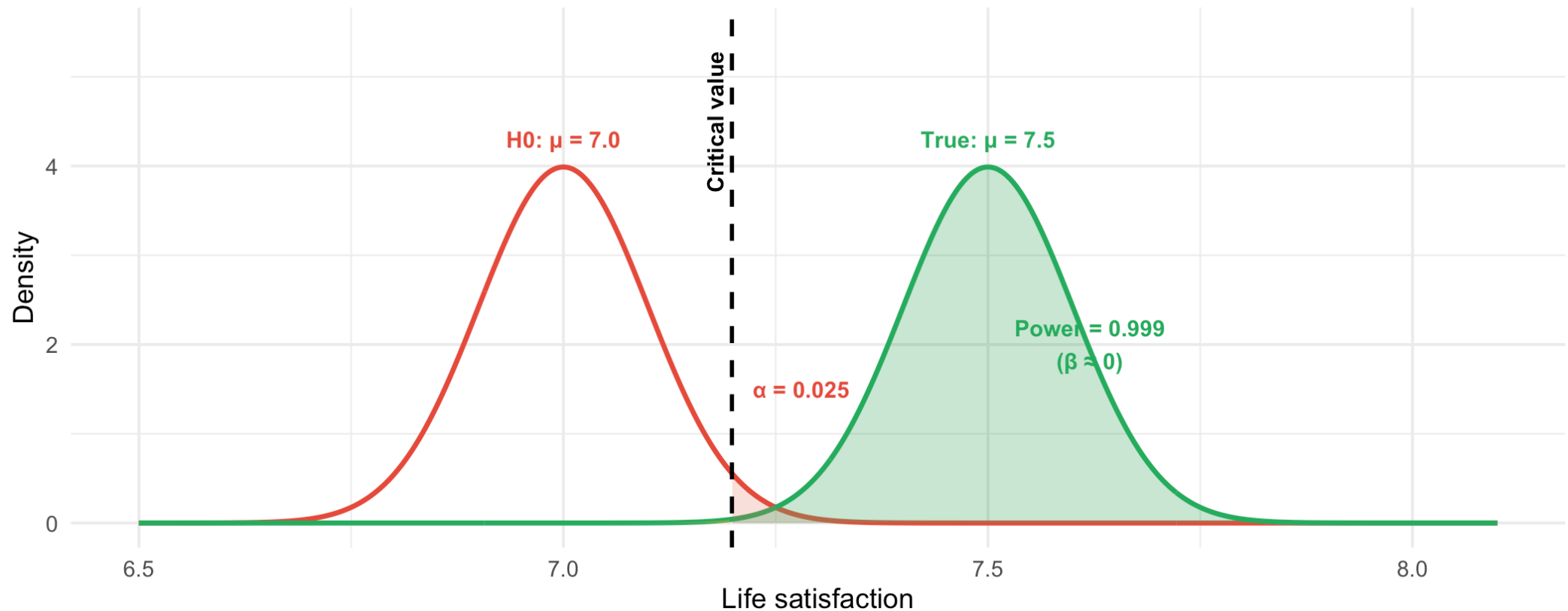
SE = 0.05 | Power = 1 (vs. 0.844 with SE = 0.1)



Power analysis: effect of larger effect size

Larger effect size: distributions further apart, power $\uparrow$

Effect = 0.5 | Power = 0.999 (vs. 0.844 with effect = 0.3)



Computing power

Power = probability of correctly rejecting H_0 when it is false

Power is the probability that $\bar{X}$ exceeds the critical value c when the *true* mean is μ_1 :

$$\text{Power} = P(\bar{X} > c \mid \mu = \mu_1), \quad c = \mu_0 + t_{\alpha/2, n-1} \cdot \frac{\sigma}{\sqrt{n}}$$

Under the true distribution $\bar{X} \sim \mathcal{N}(\mu_1, \sigma^2/n)$. Standardise:

$$\text{Power} = P\left(Z > \frac{c - \mu_1}{\sigma/\sqrt{n}}\right)$$

Substitute $c = \mu_0 + t_{\alpha/2, n-1} \cdot \sigma/\sqrt{n}$ and simplify:

$$\text{Power} = P\left(Z > t_{\alpha/2, n-1} - \frac{\mu_1 - \mu_0}{\sigma/\sqrt{n}}\right) = 1 - \Phi\left(t_{\alpha/2, n-1} - \frac{\mu_1 - \mu_0}{\sigma/\sqrt{n}}\right)$$

$t_{\alpha/2, n-1}$ replaces $z_{\alpha/2}$ because σ is unknown — but Φ (normal CDF) remains because we are standardising $\bar{X}$, not the t-statistic itself.

Computing power

As the effect ($\mu_1 - \mu_0$) or n grows, the argument of Φ shrinks $\Rightarrow$ power increases.

The three levers that increase power:

- **Larger effect size** $|\mu_1 - \mu_0|$ — the further the truth is from H_0 , the easier it is to detect
- **Larger n** — shrinks the SE, pulling the two distributions apart
- **Larger α** — moves the critical value closer to μ_0 , but increases Type I error

In R: use `pwr::pwr.t.test()` (one-sample) or `pwr::pwr.t2n.test()` (two-sample):

```
1 library(pwr)
2 pwr.t.test(d = effect / sigma, n = n, sig.level = 0.05, type = "one.sample")
```

where $d = (\mu_1 - \mu_0) / \sigma$ is Cohen's d (standardised effect size).

Computing power: example

Setup from the visualization: $\mu_0 = 7.0$, true $\mu_1 = 7.3$, $\sigma = 1.0$, $n = 100$, $\alpha = 0.05$

Analytically (manual):

```
1 mu0 <- 7.0
2 mu1 <- 7.3
3 sigma <- 1.0
4 n <- 100
5 alpha <- 0.05
6 se <- sigma / sqrt(n)
7 t_crit <- qt(1-alpha/2, df = n - 1)
8 crit <- t_crit - (mu1 - mu0)/se
9 power_an <- 1 - pnorm(crit)
```

SE = 0.1

Critical value = -1.0158

Power = 0.845

Using **pwr** (Cohen's $d = 0.3 / 1.0 = 0.3$):

```
1 library(pwr)
2 pwr.t.test(d = (mu1 - mu0) / sigma, n = n,
3            sig.level = alpha, type = "one.saml")
```

One-sample t test power calculation

```
      n = 100
      d = 0.3
sig.level = 0.05
  power = 0.8439471
alternative = two.sided
```

```
1 # the slight difference comes from normal
2 # distribution approx (left) vs t-distribution
```

Both give the same power. With $n = 100$ and a small effect ($d = 0.3$), power $\approx 80\%$ — the conventional target.

Sample size and power

How many observations do we need?

This depends on:

- The **effect size** we want to detect
- The desired **power** (typically 0.80)
- The significance level α (typically 0.05)

Sample size and power

Derivation: start from the power formula and set it equal to the desired power $1 - \beta$:

$$1 - \Phi\left(t_{\alpha/2, n-1} - \frac{|\mu_1 - \mu_0|}{\sigma/\sqrt{n}}\right) = 1 - \beta$$

Apply Φ^{-1} to both sides (recall $\Phi^{-1}(p) = z_p = -z_{1-p}$, i.e., it's **qnorm**):

$$t_{\alpha/2, n-1} - \frac{|\mu_1 - \mu_0|}{\sigma/\sqrt{n}} = -z_{1-\beta}$$

Approximate $t_{\alpha/2, n-1} \approx z_{\alpha/2}$ (valid for large n) and rearrange to isolate $\sqrt{n}$:

$$\frac{|\mu_1 - \mu_0|}{\sigma/\sqrt{n}} = z_{\alpha/2} + z_{1-\beta} \implies \sqrt{n} = \frac{(z_{\alpha/2} + z_{1-\beta}) \sigma}{|\mu_1 - \mu_0|}$$

Square both sides:

$$n = \left(\frac{z_{\alpha/2} + z_{1-\beta}}{|\mu_1 - \mu_0|/\sigma} \right)^2$$

where $|\mu_1 - \mu_0|/\sigma$ is Cohen's d , i.e., the standardised effect size (always positive)

Sample size and power

Example: What's the sample size needed to detect an effect of 0.3 points in life satisfaction ($\sigma = 2$) with 80% power at $\alpha = 0.05$? (Note: $z_{0.8} \approx 0.84$.)

Let's use the formula from the previous slide:

$$\begin{aligned} n &= \left(\frac{z_{\alpha/2} + z_{1-\beta}}{|\mu_1 - \mu_0|/\sigma} \right)^2 \\ &\approx \left(\frac{1.96 + 0.84}{0.3/2} \right)^2 \\ &\approx \left(\frac{2.8}{0.15} \right)^2 \\ &\approx 348 \end{aligned}$$

→ to estimate an effect size of at least 0.3 points (with $\sigma = 2$) with 80% power at the 5% level we need a sample size of roughly 350 individuals

Sample size and power

Example: What's the sample size needed to detect an effect of 0.3 points in life satisfaction ($\sigma = 2$) with 80% power at $\alpha = 0.05$?

R computation using the formula:

```
1 delta <- 0.3 # minimum effect to detect
2 sigma <- 2.0 # population SD (estimated from
3
4 # Critical values for  $\alpha = 0.05$  (two-sided)
5 # and power = 0.80 ( $\beta = 0.20$ )
6 z_alpha2 <- qnorm(0.975) #  $z_{\{1-\alpha/2\}} \approx 1.96$ 
7 z_alpha2
```

```
[1] 1.959964
```

```
1 z_beta <- qnorm(0.80) #  $z_{\{1-\beta\}} \approx 0.84$ 
2 z_beta
```

```
[1] 0.8416212
```

```
1 # Required n
2 n_manual <- ((z_alpha2 + z_beta) / (delta / sig
3 n_manual
```

```
[1] 348.8391
```

Verify with `power.t.test()`:

```
1 power.t.test(delta = delta, sd = sigma,
2             power = 0.80,
3             sig.level = 0.05,
4             type = "one.sample")
```

One-sample t test power calculation

```
      n = 350.7647
      delta = 0.3
      sd = 2
      sig.level = 0.05
      power = 0.8
      alternative = two.sided
```

We need **n = 349** observations.

Your turn! #7

10:00

You are planning a study on life satisfaction in France. Based on previous research, you assume $\sigma = 2.0$.

Part 1: Compute power

A previous study found $\bar{x} = 6.8$. You want to test $H_0 : \mu = 7.0$ at $\alpha = 0.05$ with $n = 80$ observations. What is the power of this test to detect the true effect?

1. Compute power manually using the formula (use `qt` and `pnorm`)
2. Verify with `power.t.test()`

Part 2: Minimum sample size

You want to design a new study to detect a difference of at least 0.4 points from $\mu_0 = 7.0$ with 80% power at $\alpha = 0.05$.

Today's lecture

The big question: How do we test whether our data provides evidence for or against a claim about the population?

1. Introduction to hypothesis testing
2. The hypothesis testing framework
3. One-sample tests
4. Two-sample tests
5. Paired tests
6. Chi-square test of independence
7. Randomization methods
8. Statistical power and sample size
- 9. Practical guidelines**

Practical guidelines

Choosing the right test

Research question	Test	Test statistic	R function
One group Mean vs. a value	One-sample t-test	$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$	<code>t.test(x, mu = ...)</code>
One group Proportion vs. a value	One-sample proportion test	$z = \frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}}$	<code>prop.test(x, n, p = ...)</code>
Two groups (indep.) Compare means	Two-sample t-test (Welch's)	$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{s_1^2/n_1 + s_2^2/n_2}}$	<code>t.test(y ~ x, data = ...)</code>
Two groups (indep.) Compare proportions	Two-sample proportion test	$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1 - \hat{p})(1/n_1 + 1/n_2)}}$	<code>prop.test(x, n)</code>
Two groups (paired) Compare means	Paired t-test	$t = \frac{\bar{d}}{s_d/\sqrt{n}}$	<code>t.test(x, y, paired = TRUE)</code>
Categorical association Test independence	Chi-square test	$\chi^2 = \sum_{i,j} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$	<code>chisq.test(table(x, y))</code>

The ASA statement on p-values

In 2016, the American Statistical Association issued [guidance](#) on proper use of p-values:

! Six principles

1. P-values can indicate how incompatible the data are with a specified statistical model
2. P-values do NOT measure the probability that the studied hypothesis is true
3. Scientific conclusions should NOT be based solely on whether a p-value passes a threshold
4. Proper inference requires full reporting and transparency
5. A p-value does NOT measure the size of an effect or importance of a result
6. By itself, a p-value does NOT provide a good measure of evidence

Bottom line: Report effect sizes, confidence intervals, and context — not just p-values!

Reporting your results

Good practice includes:

1. **Effect size**: Mean difference, odds ratio, etc.
2. **Confidence interval**: Shows precision and range of plausible values
3. **Test statistic and p-value**: Provides formal test result
4. **Sample size**: Allows assessment of power
5. **Context and interpretation**: What does it mean in plain language?

Example:

“French adults with higher education have significantly higher interpersonal trust (mean = 5.88, SD = 2.31) compared to those without higher education (mean = 5.21, SD = 2.52). The difference of 0.67 points [95% CI: 0.50, 0.85] represents a small-to-medium effect and is statistically significant ($t = 7.73$, $p < .001$).”

See you next week!